

SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY - INFORMATICS

TECHNICAL UNIVERSITY OF MUNICH

Bachelor's Thesis in Informatics

Graphical Perception in Multimodal LLMs

A Systematic Evaluation Across Visual Encoding Channels, Prompts and Models

Jakob Zöbl





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Grafische Wahrnehmung in multimodalen LLMs

Eine systematische Bewertung verschiedener visueller Kodierungskanäle, Eingabeaufforderungen und Modelle

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I confirm that this bachelor's thesis in informatics is my own work and I have documented all sources and material used.

Munich, September 18, 2026 Jakob Zöbl

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Abstract

Visualization channels differ in their ability to convey quantitative information. Early studies by Cleveland and McGill [1] established a hierarchy of channel effectiveness based on human perception. This work was later extended by Heer and Bostock [2] through large-scale crowdsourced experiments. The present thesis builds on this foundation and investigates whether Multimodal Large Language Models (MLLMs) with visual capabilities exhibit similar performance differences across visualization channels.

To this end, a replication of the experiments by Heer and Bostock [2] was conducted using several MLLMs. The original set of nine tasks was reproduced as closely as possible and extended with two additional tasks. Different prompting strategies were evaluated while remaining closely aligned with the wording of the original experiments. Based on the initial results, a second experiment was conducted to test whether the distance between compared elements affects estimation accuracy or whether visual distractors are the dominant factor.

The results show a clear distinction between geometric and color-based channels. Position, length, angle and other geometric channels yielded similar performance across the tested models. In contrast, channels based on luminance and saturation resulted in substantially larger estimation errors. This pattern is consistent with findings from human perception studies, although no clear ranking emerged among the geometric channels. The second experiment found no significant effect of distance on estimation accuracy. Instead, the results suggest that visual distractors are the primary source of error. This interpretation is supported by performance differences between tasks that introduce additional distractors. Overall, the findings indicate that MLLMs exhibit channel-dependent performance patterns that resemble those observed in human graphical perception. Mainly, they prefer geometric over color based channels and that their errors are driven predominantly by the introduction of distractor elements rather than the distance between compared elements.

Kurzfassung

Visualisierungskanäle unterscheiden sich in ihrer Fähigkeit, quantitative Informationen zu vermitteln. Frühe Studien von Cleveland und McGill [1] legten auf der Grundlage der menschlichen Wahrnehmung eine Hierarchie der Kanaleffektivität fest. Diese Arbeit wurde später von Heer und Bostock [2] durch groß angelegte Crowdsourcing-Experimente erweitert. Die vorliegende Arbeit baut auf dieser Grundlage auf und untersucht, ob MLLMs mit visuellen Fähigkeiten über verschiedene Visualisierungskanäle hinweg ähnliche Leistungsunterschiede aufweisen.

Zu diesem Zweck wurde eine Replikation der Experimente von Heer und Bostock [2] unter Verwendung mehrerer MLLMs durchgeführt. Der ursprüngliche Satz von neun Aufgaben wurde so genau wie möglich nachgebildet und um zwei zusätzliche Aufgaben erweitert. Es wurden verschiedene Prompt-Strategien evaluiert, wobei man sich eng an den Wortlaut der ursprünglichen Experimente hielt. Auf der Grundlage der ersten Ergebnisse wurde ein zweites Experiment durchgeführt, um zu testen, ob der Abstand zwischen den verglichenen Elementen die Schätzgenauigkeit beeinflusst oder ob visuelle Ablenkungsfaktoren der dominierende Faktor sind.

Die Ergebnisse zeigen eine klare Unterscheidung zwischen geometrischen und farbbasierten Kanälen. Position, Länge, Winkel und andere geometrische Kanäle führten bei allen getesteten Modellen zu einer ähnlichen Leistung. Im Gegensatz dazu führten Kanäle, die auf Luminanz und Sättigung basierten, zu wesentlich größeren Schätzfehlern. Dieses Muster stimmt mit den Ergebnissen aus Studien zur menschlichen Wahrnehmung überein, obwohl sich unter den geometrischen Kanälen keine klare Rangfolge ergab. Das zweite Experiment ergab keinen signifikanten Einfluss des Abstands auf die Schätzgenauigkeit. Stattdessen deuten die Ergebnisse darauf hin, dass visuelle Ablenkungsreize die Hauptfehlerquelle darstellen. Diese Interpretation wird durch Leistungsunterschiede zwischen Aufgaben gestützt, bei denen zusätzliche Ablenkungsreize eingeführt wurden. Insgesamt deuten die Ergebnisse darauf hin, dass MLLMs kanalabhängige Leistungsmuster aufweisen, die denen ähneln, die bei der grafischen Wahrnehmung des Menschen beobachtet werden. Vor allem bevorzugen sie geometrische gegenüber farbbasierten Kanälen und ihre Fehler werden überwiegend durch die Einführung von Ablenkungselementen und nicht durch den Abstand zwischen den verglichenen Elementen verursacht.

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1. Introduction

Graphical data visualization is a foundation of contemporary data analysis, functioning as an essential part of communication, exploration and decision making in science, business and politics. At its core, the effectiveness of a visualization is measured by how well it maps data to visual properties such as position or length. For example, this mapping could mean that after an election, each party's share of the vote (in percentage) will be displayed in a bar chart (channel: position on a common scale) to convey quantitative information (the voting outcome) more intuitively to the audience. To ensure that humans can easily read and accurately interpret the displayed information, charts are designed using established principles of visual perception [3].

The formalization of these principles has its roots in earlier works such as Jacques Bertin's *Semiology of Graphics* [4], which explored visual variables primarily in the context of maps. It was later empirically grounded by the research of Cleveland and McGill [1]. They worked to establishing a now widely accepted hierarchy of visual encoding channels. By empirically testing human accuracy in extracting quantitative information from various stimulus images, they demonstrated that humans exhibit distinct perceptual biases when judging visual encoding channels. These rankings changed the field of information visualization by shifting the focus from aesthetic craft to a studied science. The resulting hierarchy has since been extended to include additional channels, as well as categorical and ordinal information. It influenced decades of best practices and formed the basis for automated chart generation systems [5].

However, the rapid development of Artificial Intelligence (AI) is already changing how visual data is processed and interpreted. Specifically, the increase in the capability and popularity of MLLMs, which can interpret, reason about and generate visuals without specialized training, has made them another important consumer of visual data alongside humans. They are increasingly embedded in a variety of modern tools that read, explain, design charts and even serve as intelligent agents without human involvement. This fundamental change in how humans interact with visual data through MLLMs as intermediaries raises an important new question. Do these models share the same perceptual biases as humans, or do they exhibit their own, different perceptual hierarchies?

Understanding the graphical perception capabilities of MLLMs is critical for their safe and effective deployment in a world that is rapidly moving towards increasingly AI-driven workflows. If an AI system ranks visual channels differently than a human, it is more likely to misread a chart that was explicitly created with human-centered design principles in mind. Consequently, a better knowledge of these differences could call for a rethinking of our long-standing design principles, to transition from purely human-centric standards to "AI-oriented" or hybrid visualizations that optimize readability for both humans and MLLMs.

Conversely, the differences could also require fine-tuning future MLLMs to align more closely with the human baselines.

While the evaluation of MLLMs is a growing field, research into their fundamental perceptual biases remains limited. For instance, prior research has investigated how visual data is processed by various modern AI architectures, such as Vision Transformers (ViTs) and Convolutional Neural Networks (CNNs) [6]. It has assessed their ability to detect and correct formal chart design violations [7] and benchmarked their accuracy on low-level data extraction and reasoning tasks [8]. Yet, no research has proposed a comprehensive and clear visual encoding channel hierarchy for current state-of-the-art MLLMs. This thesis aims to help fill this gap in the literature.

1.1. Research Aim and Contribution

By creating an open-source experimental setup, this work aims to recreate, adapt and expand the classical graphical perception experiments originally conducted by Cleveland and McGill [1] and later by Heer and Bostock [2]. Through this systematic study across various state-of-the-art models and prompting strategies, this thesis aims to establish a perceptual hierarchy of visual encoding channels for modern MLLMs, similar to what has been done for humans.

1.1.1. Main Research Question - RQ0

Do MLLMs exhibit a generalized, consistent and measurable perceptual hierarchy across a given set of visual encoding channels for quantitative information?

The core question of this thesis stems from foundational studies of human perception that have established our current knowledge of visual encoding channels. While modern hierarchies, such as those synthesized by Munzner [9], incorporate results from various works, this thesis focuses on evaluating a core subset of these channels. It aims to determine if similar patterns can be found in modern MLLMs by recreating and building on classical experiments such as the crowdsourced experiment by Heer and Bostock [2]. Such rankings could become vital to ensuring that visualizations are designed and interpreted appropriately as we delegate more and more tasks to MLLMs.

1.1.2. Sub-Question - RQ1

To what extent does prompt design (Chain-of-Thought (CoT), phrasing, etc.) influence MLLM performance on low-level graphical perception tasks and alter the relative ranking of visual encoding channels?

Previous studies have shown that Large Language Models (LLMs) can be highly sensitive to the prompts they are given. This question aims to separate a model's intrinsic visual perception capabilities from its instruction bias. By testing different prompt designs, we can begin to understand how reliant perceptual accuracy is on the input prompt.

1.1.3. Sub-Question – RQ2

How does LLM low-level graphical perception accuracy compare to human performance, both in terms of absolute estimation error and the resulting ordering of visual encoding channels?

Today, data visualizations and charts are intrinsically designed for human observers, informed by established design principles. This question directly compares AI performance with human results to identify potential gaps or overlaps.

1.1.4. Sub-Question – RQ3

To what extent does the capability tier or architecture of an MLLM affect its graphical perception performance?

Current state-of-the-art MLLMs vary in terms of architecture, ranging from dense models to Mixture of Experts (MoE), total number of parameters and context window size, for example. This question aims to answer if and how these differences affect graphical perception accuracy. Answering this could help anticipate how future models might perform on visual tasks, depending on their design.

1.2. Thesis Structure

This thesis is structured as follows. It starts by providing the necessary background on graphical perception, visual encoding channels and MLLMs. Following this, the related work section discusses prior human perception studies and recent research on visual processing in LLMs and, more broadly, in machine learning. The overarching methodology and the exact experimental setup are then explained in a thorough and comprehensive chapter, outlining details such as image generation, comparison logic, metrics and prompt configurations used to evaluate the models. Finally, the thesis presents the results and offers a thorough discussion of their implications, before ending with a conclusion and an outlook on future research.

2. Background and Related Work

This chapter establishes the theoretical and historical foundations necessary to understand the experiments conducted in this work. It presents foundational literature on human graphical perception. It then introduces the architecture and mechanics of MLLMs, detailing how they handle visual information. Finally, it surveys recent related work on AI visual understanding and prompt engineering to put this work's contribution into context within the wider field of machine perception.

2.1. Human Graphical Perception

This section reviews the foundational literature on human graphical perception, focusing entirely on how biological and cognitive constraints shape the way people read charts. It introduces the core theories of visual encoding and pre-attentive processing that explain why certain visual variables are more effective than others. It also covers the seminal experiments that empirically ranked these encoding channels, establishing the human baseline that this thesis aims to replicate with machine learning models.

2.1.1. Foundations of Graphical Perception and Visual Information Encoding

The formalization of visual encoding principles can be traced back to Jacques Bertin's *Semiology of Graphics* [4], which laid out a theoretical framework for mapping data to visual variables. Later, Colin Ware [3] provided the cognitive reasoning behind these mappings. He argued that pre-attentive visual processing gives certain comparisons, such as position and length, a fundamental biological advantage in speed and accuracy. Graphical perception, as Cleveland and McGill [1] introduced it, describes the process of visually decoding information within graphical mediums. Their seminal experiments introduced multiple *elementary perceptual tasks*, based on different *visual encoding channels*. These channels are different ways to display the same underlying information visually. For example, Cleveland and McGill used tasks based on position on a common scale (e.g. bar charts), length (e.g. stacked bar charts), angle (e.g. pie charts) and area (e.g. tree maps). Examples of the mentioned charts and the tasks based on them are shown in Figure 3.2. The channels got empirically ranked by the accuracy with which humans perform the tasks based on them, leading to our modern perceptual hierarchy of visual encoding channels. These rankings have since informed chart-design best practices [1, 9] and remain important to visualization research.

As this hierarchy establishes, the effectiveness of a visualization depends on how data attributes are mapped to visual channels. According to Munzner [9], data attributes can be classified as *quantitative* (numerical values with a meaningful underlying scale, e.g. height),

ordinal (ordered categories without sensible ratios, e.g. low, medium, high), or *nominal* (unordered categories, e.g. names). The visual encoding channels examined in this work (position, length, angle, area, color luminance, color saturation) are only examined in the context of quantitative information. Their relative accuracy for humans follows a well-established ranking. Position on a common scale allows for the most precise judgments, followed by length, angle, area and finally color-related channels [1, 2, 9].

The stimuli used for the experiment in this thesis recreate the designs of Cleveland and McGill [1] and Heer and Bostock [2]. Detailed descriptions of each task type and example stimuli are provided in Section 3.2 and Figure 3.2.

2.1.2. Human Perception Experiments and Channel Hierarchies

Cleveland and McGill [1] introduced the term *graphical perception* to describe the visual decoding of quantitative information from charts. In their experiments, human participants estimated proportional differences between two marked values across seven task types, yielding an empirically grounded ranking for a specific set of visual encoding channels used in their studies. The resulting hierarchy places position on a common scale at the top. It quickly became one of the most cited results in information visualization. Its reach extended well beyond the perception lab. Mackinlay [5] built directly on these rankings to formalize the automated design of graphical presentations, encoding channel effectiveness as a rule-based ordering to select the most appropriate visual mapping for a given data attribute. The broader perceptual theory that explains *why* these rankings hold was developed by Ware [3], who situated graphical perception within the architecture of the human visual system. He argued that early, pre-attentive visual processing gives certain comparisons, especially those involving position and length, a fundamental speed and accuracy advantage over others that require deliberate, attention-driven effort. Munzner [9] later brought much of this accumulated knowledge together into a practical design framework. She refined the channel rankings for quantitative, ordinal and nominal data types, making them directly applicable for visualization practitioners.

Heer and Bostock [2] replicated and extended the experiments from Cleveland and McGill using crowdsourcing platforms to increase scale. They confirmed the original channel ranking and added new tasks for rectangular area judgments, for example. Beyond the replication itself, their use of Amazon Mechanical Turk demonstrated that statistically reliable perceptual data could be collected at scale without a laboratory setting. This has since made larger participant pools feasible for this kind of research. Their dataset also covers rectangular area stimuli of the kind used in treemap-style layouts [10] and provides the primary human baseline against which subsequent machine perception work, including this thesis, compares.

More recently, Guo et al. [11] began exploring these exact same tasks in the context of MLLMs. However, their study primarily focused on a single model (GPT-4o-mini). In contrast, this thesis systematically evaluates these foundational tasks across a wide range of model tiers and architectures to determine whether these human biases generalize across the AI landscape.

2.2. Multimodal Large Language Models

This section provides the necessary background on (M)LLMs. It covers their underlying architecture, general properties as well as their visual capabilities. It begins by explaining the fundamental mechanics of modern vision-language models, including how they discretize images. It then puts this work into the context of prior research on how these models perform on graphical perception tasks and chart comprehension. Lastly, it highlights the significant role that prompt engineering plays in their performance.

2.2.1. MLLM Architecture and Properties

State-of-the-art LLMs are fundamentally based on two different underlying language model architectures. *Dense* models, which activate all parameters for every input token and *MoE* architectures, which route each token to a small subset of specialized expert sub-networks. Those activate only a portion of the total parameters for any given input token. This architectural choice primarily affects inference cost, memory footprint and scaling behavior.

An MLLM extends a standard LLM by including a vision encoder module. This enables the model to understand and reason over both visual and textual inputs [12]. While regular language models exclusively process token sequences directly put in via a prompt, MLLMs map an input image into a sequence of *visual tokens*. They are then also projected into a given embedding space. The combined sequence of visual and textual tokens is then processed in the same way as a sequence of regular text tokens. This is done by a transformer decoder, which generates responses token-by-token autoregressively using causal self-attention and cross-modal attention.

The vision encoder in most modern MLLMs is based on the ViT architecture [13]. A ViT splits an image into fixed-size chunks, then embeds each chunk individually and processes the resulting sequence through transformer layers. The output embeddings are passed through a learned projection layer to align them with the language model's textual token space. This architecture has been adopted across a wide range of vision-language models and serves as the standard visual paradigm for current MLLMs. One major drawback of this patch-based discretization is that it inherently disrupts fine-grained spatial relationships. Consequently, models may struggle with precise localization, which is a crucial capability for accurately reading charts.

Two additional important properties that constrain how MLLMs process given information, visual and textual alike, are the *context window* and the *total parameter count*. The context window defines the maximum number of total input tokens that a model can handle in a single pass. This includes both text and visual tokens. Its size therefore limits the resolution and number of visual stimuli or regular text that a model can jointly reason about. Furthermore, the total number of parameters in a model constrains its ability to reason about complex inputs. Larger models tend to be more capable, but are also more expensive to run and train according to the general scaling laws of neural networks [14].

2.2.2. Vision Models and Graphical Perception

Before MLLMs became the dominant paradigm in machine vision research, earlier work had already begun to ask how non-language-based visual architectures cope with the specific demands of reading charts. Poonam et al. [6] test multiple ViT architectures against CNNs and humans on low-level graphical perception tasks derived directly from the original Cleveland and McGill experiments. Despite strong performance on general vision benchmarks, ViTs underperform compared to both humans and CNNs in this test. This suggests that strong general vision performance does not automatically translate into human-like graphical perception. This finding has direct implications for MLLM research, given that the ViT architecture [13] forms the visual backbone of virtually all current MLLMs. In other words, the chart-reading weakness that Poonam et al. identify in isolated ViT models might be carried into every MLLM that inherits this backbone. This leaves open the question of whether the additional language-model layers compensate for this structural limitation or merely work around it in prompt-dependent ways.

2.2.3. Prior Work on Visualization Understanding in Multimodal LLMs

The closest prior work to this paper is Guo et al. [11], who evaluate GPT-4o-mini in a zero-shot setting on the first seven Cleveland and McGill and Heer and Bostock tasks. They recreate the original graphics, prompt the model and compare results to human data. They find that GPT-4o-mini can show rankings comparable to humans under certain prompt templates (e.g. with explicit color references and requested CoT explanations), but performance drops and rankings shift noticeably when these cues are removed. Crucially, their study uses only a single model, which limits the generalizability of the findings to the broader field of MLLMs. Whether the observed behavior is characteristic of modern MLLMs in general or whether it is specific to GPT-4o-mini's training and architecture is still an open question.

Other work analyzes visualization literacy and chart comprehension from complementary angles. Dong and Crisan [15] evaluated a range of models, including GPT-4o, GPT-3.5, Gemini and GPT-5.4, on chart generation and understanding tasks. They found systematic differences across model families that do not reduce cleanly to a single underlying capability dimension. A subsequent analysis by the same authors [16] zoomed in further and identified specific thresholds in visualization understanding across modern MLLMs and reports failure modes such as difficulties with color-intensive charts. Han et al. [8] introduce ChartInsights, a large-scale benchmark for low-level chart question answering and show that even frontier models at that time, such as GPT-4o, achieve only around 69% accuracy. This result underlines how consistently visual chart information gets misread, even by capable models. Sinnona et al. [7] take a different angle, testing models to detect and correct violations of visualization design principles. It finds that models are better at fixing violations they have been shown than at independently identifying them. This asymmetry hints at a gap between pattern-level procedural knowledge, absorbed during pretraining and a more flexible conceptual grasp of what makes a chart legible. Work on bias in large-scale visual training datasets [17] adds another layer to this picture. The results suggest that the statistical composition of pretraining

corpora and especially the relative scarcity of structured chart stimuli compared to natural photographic images may introduce systematic biases. These biases are difficult to disentangle from genuine model capability.

2.2.4. Prompting Methodologies

When benchmarking MLLMs, the input instruction given to a model in the prompt often defines task-specific guidance. In *zero-shot* prompting, as used in this thesis, the model receives only a task description and the stimulus, with no prior examples. This strategy aims to isolate the model's intrinsic, pre-trained perceptual and reasoning abilities from its capacity for in-context learning. In *few-shot* prompting on the other hand, one or more example input-output pairs are provided in the prompt. The idea behind this is to allow the model to observe and learn from the task patterns presented in the examples.

Modern prompt engineering includes several techniques, including *role-playing*, *CoT* and *duplication* prompting [18]. Role-playing instructs the model to adopt a specific persona (e.g. "expert data analyst") to help it focus on a specific field and view a problem from a particular perspective.

CoT prompting is a completely separate and independent technique where the prompt explicitly instructs the model to generate intermediate reasoning steps before providing a final answer [19]. This is done by instructing the model to verbalize its analytical process rather than answering immediately. Step-by-step thinking has been shown to significantly improve performance on complex tasks [19]. It may help models in the visual domain to more accurately reason over the provided input images [11].

Lastly, duplication prompting involves systematically repeating core instructions within the same prompt [20]. This helps the model to reinforce constraints and prevent it from "losing" context across attention heads.

This work evaluates these varied prompting conditions in a zero-shot setting to comprehensively isolate and analyze the effect of instruction design on low-level graphical perception across different capability tiers. We apply few-shot prompting only to the output pattern, showing the LLM exactly how we expect the output to be formatted. More on this in section 3.4.

A thread running through almost all of the studies above is that prompt formulation strongly mediates how MLLMs perform on visual tasks. Liu et al. [12] showed with LLaVA that instruction-tuned vision-language models can follow complex multimodal instructions with reasonable fidelity, but also that the quality and consistency of responses are sensitive to how instructions are worded in ways that are not always predictable. Guo et al. [11] directly demonstrated this sensitivity for graphical perception tasks. The presence or absence of CoT instructions shifts not only absolute accuracy but also the relative ranking of visual encoding channels. Whether this reflects a genuine improvement in visual perception or is mainly a change in the model's output toward more conservative numerical estimates is a question that isolated single-model evaluations struggle to resolve.

2.3. Research Gap and Thesis Contribution

Overall, prior research shows that MLLMs can partially reproduce human graphical perception behavior, but their performance is highly sensitive to task design, stimuli and prompting. What the literature still lacks is a systematic comparison spanning multiple models with different architectures and capability tiers, evaluated under consistent experimental conditions using the same stimuli and question formats that underlie the existing human baselines.

By replicating Heer and Bostock's scalable crowdsourcing tasks, this thesis establishes a robust, human-anchored baseline for MLLM evaluation. Applying these exact same stimuli across a diverse array of language models and prompting strategies fills the gaps left by previous single-model studies. This systematic approach allows us to determine whether human perceptual biases are universal across modern MLLMs or merely artifacts of specific prompt structures and model architectures.

3. Methodology and Experimental Setup

This chapter details the overarching methodology and the exact experimental setup used to evaluate the graphical perception capabilities of modern MLLMs. At a high level, the experiment consists of three primary stages, as illustrated in Figure 3.1. First, a large dataset of synthetic chart images is programmatically generated. These images cover a wide variety of visual encoding channels, such as position, length and area and are designed to test fundamental perceptual biases. Second, these images are systematically fed into a diverse selection of MLLMs via API endpoints. The models are queried using a range of prompting strategies to isolate their visual reasoning abilities from their sensitivity to prompts. Finally, the models' numerical estimates are parsed and statistically evaluated to assess how accurately they extracted quantitative information from the charts, enabling us to build a comprehensive perceptual hierarchy.

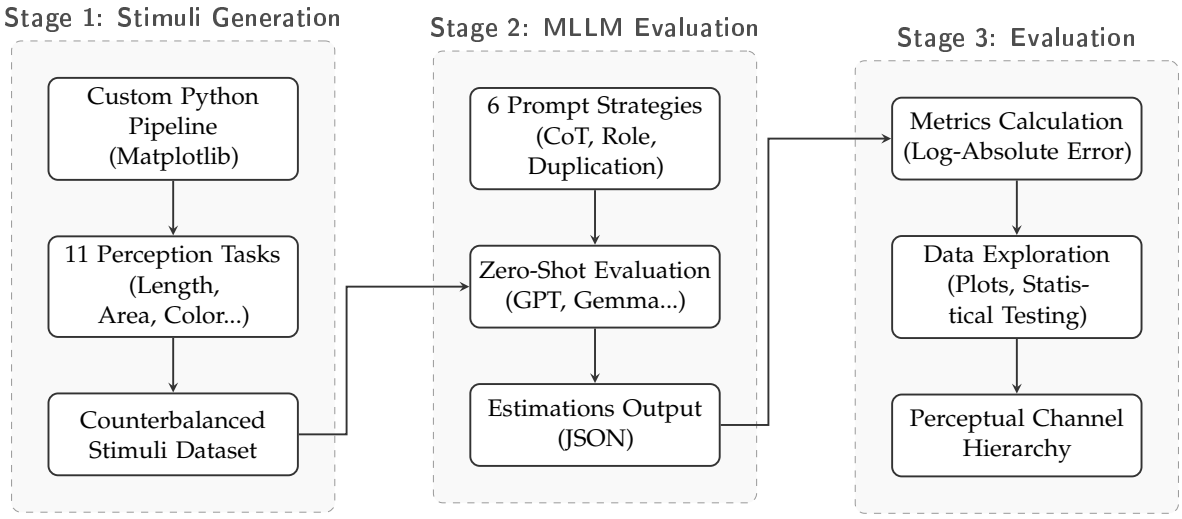


Figure 3.1.: High-level overview of the experimental pipeline: (1) generation of graphical stimuli across various visual channels, (2) evaluation of multiple MLLMs using diverse prompting strategies and (3) parsing and statistical analysis of the estimations to establish a perceptual hierarchy.

3.1. Study Design

The overarching goal of this thesis is to systematically evaluate the graphical perception capabilities of MLLMs and analyze whether they share the same foundational visual processing

biases observed in humans. To achieve this, the study design is structured as a near direct analog and expansion of the classical experiments conducted by Cleveland and McGill [1] and later crowdsourced by Heer and Bostock [2]. However, adapting an experiment originally intended for human subjects to MLLMs requires multiple methodological adjustments. Unlike humans, who process visual scenes continuously through biological optical systems, MLLMs process images by discretizing them into visual tokens via a ViT and projecting them into a language decoder's embedding space. This fundamental difference leads to a series of changes to thoroughly control all study parameters, such as prompt phrasing and cross-model generalization.

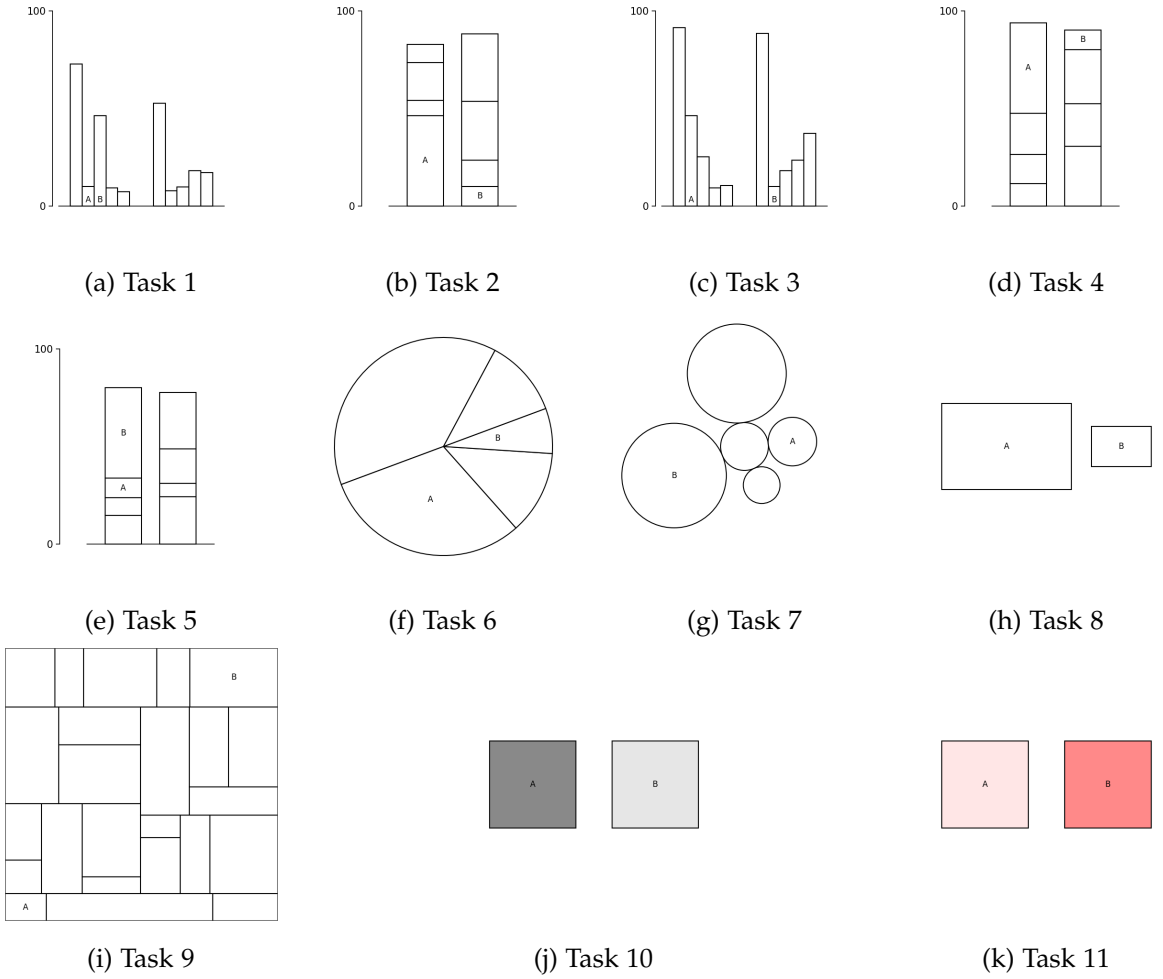
Instead of human participants seated in front of a paper and pencil or completing tasks via crowdsourcing platforms like Amazon Mechanical Turk, the subjects of this study are a diverse set of MLLMs accessed via API endpoints. The models are presented with graphical stimuli generated to closely match those used by Heer and Bostock [2] (see Section 3.2). The tasks required the models to perform elementary perceptual tasks, specifically, identifying the larger of two marked graphical elements and estimating the percentage ratio between them. We utilize a zero-shot inference approach throughout all evaluations. This ensures that we are probing the models' intrinsic, pre-trained visual capabilities rather than their ability to learn a specific task in-context through few-shot examples. In-context learning could artificially inflate performance.

To capture a comprehensive picture of model behavior, we evaluate a wide spectrum of MLLMs, ranging from massive, proprietary frontier models (GPT-5.4 and Gemini 3.1 Pro) to smaller, open-weight and free-tier models (such as Gemma 4 and Nemotron). They also represent a wide range of models, differing in their architecture, parameter count and alignment training. To ensure more reliable results, we test each model using a multi-prompt strategy to isolate visual capabilities from prompt sensitivity.

3.2. Task and Stimuli Design

To ensure the results of this study can be meaningfully compared with the established literature on human graphical perception, the generated stimuli need to closely mirror the visuals used by Heer and Bostock [2], which were themselves built on the original five task types by Cleveland and McGill [1]. We decided to retain the exact same nine elementary graphical perception tasks and keep them grouped by their respective visual encoding channels [9]. Additionally, we introduced two new task types not presented in either of the foundational studies. Task types ten and eleven feature color channels, color luminance and color saturation, respectively. With this addition, we extend the hierarchy by two more visual encoding channels, which are typically ranked very low in human perception [9] but could score highly for MLLMs, as previous research suggests they are highly sensitive to color [8, 11].

That being said, running a perception experiment designed for humans on machine vision systems required a few practical adjustments. For instance, in the original study, the elements participants were asked to compare in tasks one through five were simply marked with small



(a) Task 1 (b) Task 2 (c) Task 3 (d) Task 4

(e) Task 5 (f) Task 6 (g) Task 7 (h) Task 8

(i) Task 9 (j) Task 10 (k) Task 11

Figure 3.2.: Representative examples of stimuli for all eleven graphical perception tasks. Tasks 1–3 test position on a common scale using grouped and stacked bar charts. Tasks 4–5 test length via stacked bar variations. Task 6 tests angle/tilt with a pie chart. Task 7 tests the circular area with a bubble chart. Tasks 8–9 test rectangular area using isolated rectangles and treemaps. Tasks 10–11 test color luminance and saturation using equally sized squares. Elements marked A and B are the targets for proportional estimation.

dots. While humans can easily explain which marked elements is larger, a programmatic evaluation requires concrete values. We found this to be a potential stumbling block for MLLMs. To address this, the dots were replaced with explicit *A* and *B* text labels. This makes the visuals, the instructions and, perhaps more importantly, the output much more explicit to parse and it allows us to reference the elements clearly within the text prompts and the model responses. Interestingly, Heer and Bostock already used similar alphanumeric labels for their later tasks (six through nine), so this change simply standardizes the labeling approach across

the entire experiment. Adopting this strategy also aligns our stimuli with recent MLLM evaluation frameworks, such as the one used by Guo et al. [11].

The charts themselves were generated programmatically using custom Python scripts built around the Matplotlib library. Building the images from scratch rather than relying on screenshots from older papers was essential. It gave us complete control over the resolution, element sizing and spacing, while ensuring the images were free of unwanted compression artifacts. The following subsections break down exactly how these images were constructed, detailing the underlying formulas, layout rules and specific constraints applied to each task type. A deeper dive into why these specific mathematical constraints (3.1) and (3.2) were chosen in the first place can be found in the original work by Cleveland and McGill [1] and its replication by Heer and Bostock [2].

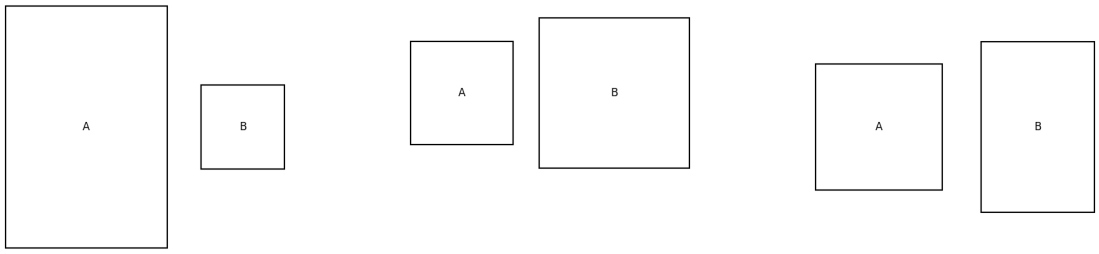
3.2.1. Channel Selection

To ensure clean comparability with prior work, especially Heer and Bostock [2], task types 1 to 9 are designed to match those in their experiment. Tasks 10 and 11 are additions that extend the results with color-based channels and were not present in the prior work of Heer and Bostock, Cleveland and McGill, or Guo et al. [2, 1, 11]. This means the channels also stayed the same. In selecting these specific tasks and channels, the primary aim was to enable direct comparison with foundational studies while also allowing for generalizability to a broader set of visual encoding strategies commonly used in data visualization. Nevertheless, it is important to acknowledge that not all chart types strictly fall into a single visualization channel. For example, a bar chart as seen in Figure 3.2a uses length, position on a common scale and rectangular area all at the same time. Similarly, a pie chart uses angle, arc length and area. Cleveland and McGill [1] make a similar point and argue that for ease of interpretability of results, it is important to only use each chart's *primary* encoding channel. The same approach is adopted here. Consequently, while this selection prioritizes methodological consistency and comparability, it remains sufficiently representative for generalizing findings to a wide range of perceptual tasks in graphical data analysis.

In addition to the nine previous task types, we introduced task types 10 and 11 to evaluate the visual channels for color luminance and color saturation. Traditionally, color-based channels are ranked significantly lower in terms of effectiveness for encoding quantitative variables within human graphical perception hierarchies [1, 9]. However, evaluating models on these tasks allows us to determine whether this classical human assumption holds for artificial vision systems, or whether their architectural characteristics lead to different performance profiles when processing non-spatial visual properties. Specifically, these two channels were selected because they typically rank the highest for human perception biases among color based channels.

3.2.2. General Image Specifications

The images were all generated in a resolution of 600×600 pixels and in the PNG format. To adhere as closely as possible to the foundational work of Heer and Bostock [2] and Cleveland



(a) Ratio: 0.18 (easiest)

(b) Ratio: 0.46 (medium difficulty)

(c) Ratio: 0.83 (hardest)

Figure 3.3.: Examples of Task 8 (Rectangular Area) stimuli showcasing different estimation difficulties with varying true percentage ratios.

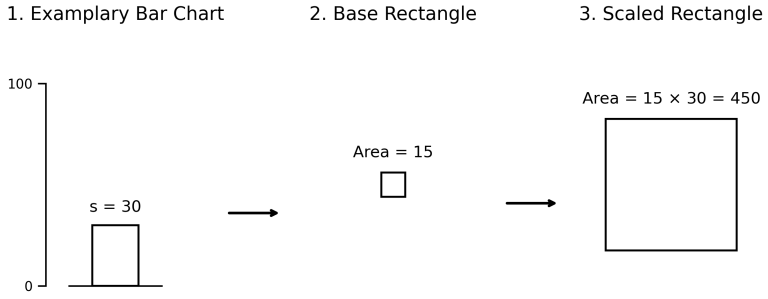


Figure 3.4.: Visual explanation of the area scaling for Tasks 8 and 9. A given s_i value is multiplied by a base area of 15 to calculate the final pixel area of the rectangle. Other chart types can use the s_i value directly as displayed in the exemplary bar chart.

and McGill [1], all charts are shown on a white background using a grayscale scheme with black as the darkest and white as the lightest. Additional color information was only given for charts 10 and 11, which test color luminance and saturation, respectively.

For all task types, the following formula was used to generate the underlying data values for the two specific elements under evaluation:

$$\begin{aligned} s_i &= 10 \cdot 10^{(i-1)/12} \\ i &\in 1, \ldots, 10 \end{aligned} \tag{3.1}$$

The formula gives us ten values on a logarithmic scale ranging from 10 to 56.2. These values are used to create a list of 45 unique value pairs with nine different ratios between them, ranging from 0.18 to 0.83, representing different difficulties of estimation tasks [1]

(see Figure 3.3). The ratios are direct results of how Weber's law affects human perception. Instead of perceiving changes in absolute values on a linear scale, the human visual system operates logarithmically, meaning that a *constant ratio* between two values is perceived as a *constant difference*. According to Cleveland and McGill [1], these ratios directly dictate the difficulty of the estimation task. For example, estimating a ratio of 0.5 (where one element is exactly half the other) is generally easier for both humans and machines than estimating a highly specific ratio like 0.83.

Where the other nine task types all used the results of 3.1 directly as absolute values, task types eight and nine used them as multipliers for the area of the rectangles with the base set to 15. For example, an s_i value of 30 generates a rectangle with a total area of $15 \times 30 = 450$ (see Figure 3.4). Multiplying all s_i values by a constant factor ensures that the ratio between any pair and the log scale properties are preserved, leading to consistent comparisons across different tasks [1].

For task types eight and nine, the rectangles are additionally required to have aspect ratios that are given by the Cartesian product of the set

$$\{2/3, 1, 3/2\} \tag{3.2}$$

with itself, resulting in nine different pairs of aspect ratios. This perfectly fits the grouping methodology, which puts nine pairs in each group, allowing us to assign a unique combination of aspect ratios to each pair in a group. The exact ordering of the aspect ratios remains the same across all groups of value pairs to ensure better comparability of experimental results. The used sequence was randomly generated and is shown in Appendix A.3.

The other quantitative values shown in the stimulus images were generated randomly with different limitations in place. All constraints are drawn directly from the work of Heer and Bostock [2] and Cleveland and McGill [1]. Detailed per-task restrictions and chart constructions are provided in the following subsections.

3.2.3. Task Specific Constraints

Task 1
Channel: (vertical) position on common scale.
Chart type: bar chart
Task type 1, as shown in Figure 3.5, uses a vertical bar chart with a total of ten equally wide rectangles of varying height. The ten bars are split into two groups, with five elements in each. The leftmost bar in each group represents the total length of the four other bars in the group. The marked elements for comparison are always the second and third bars from the left in the left group. The right group is not relevant for the comparison task. The random distractor values are drawn from a uniform distribution.

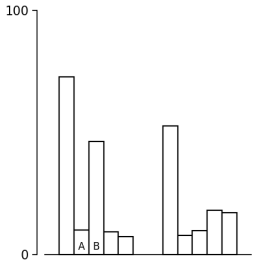


Figure 3.5.: Example stimulus for Task 1.

Task 2
Channel: (vertical) position on common scale.
Chart type: stacked bar chart
The second task, illustrated in Figure 3.6, uses a stacked bar chart with two separate bars. Each bar consists of four differently sized elements and both are placed side by side on a common scale. The two bottom elements are marked for comparison. The random distractor values are again drawn from a uniform distribution and do not affect the task.

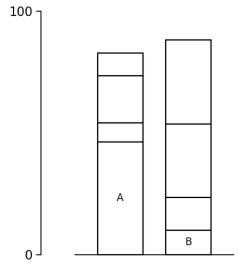


Figure 3.6.: Example stimulus for Task 2.

Task 3
Channel: (vertical) position on common scale.
Chart type: bar chart
For the third type (Figure 3.7), we again use a vertical bar chart with the same constraints as in task type one. Here, the two marked bars for comparison are always the second one from the left in each group.

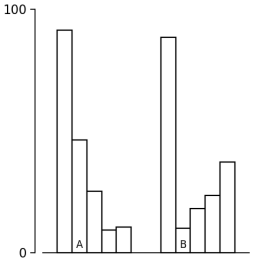


Figure 3.7.: Example stimulus for Task 3.

Task 4
Channel: length
Chart type: stacked bar chart
In task type 4 (see Figure 3.8), the used channel changes to length. This is achieved using the same setup as in task 2, meaning a stacked bar chart with 4 elements per bar on a common scale. This time, the two marked elements are the top elements of each bar instead of the bottom ones. Additionally, the distractor values are not allowed to align at the top or bottom of the second element from the top, since this would again put the two top elements on a common scale.

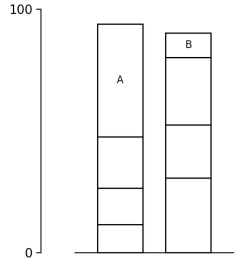


Figure 3.8.: Example stimulus for Task 4.

Task 5
Channel: length
Chart type: stacked bar chart
Task 5, depicted in Figure 3.9, is similar to task 4 as it uses a stacked bar chart for length comparison. Instead of marking the top elements of each of the two stacked bars, it marks the two top elements of the left bar, completely ignoring the right one.

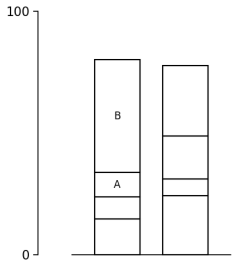


Figure 3.9.: Example stimulus for Task 5.

Task 6
Channel: angle/tilt
Chart type: pie chart
The sixth task (Figure 3.10) uses a pie chart with five sectors. The marked ones are not allowed to be contiguous, meaning they must be separated by at least one distractor value. The random values are chosen between 3.0 and 39.0, like in Heer and Bostock [2].

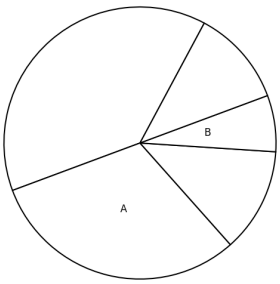


Figure 3.10.: Example stimulus for Task 6.

Task 7
Channel: (circular) area
Chart type: bubble chart
In the first of the three area judgment tasks (Figure 3.11), the stimuli consist of 5 contiguous circles with no intersections but touching borders. The s_i values are explicitly used for the area and not for the radius of the circles. The distractor elements are again drawn from a uniform distribution.

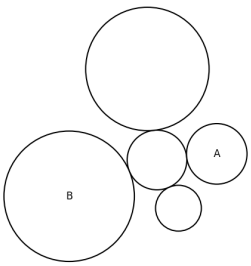


Figure 3.11.: Example stimulus for Task 7.

Task 8
Channel: (rectangular) area
Chart type: squares
Task type eight, as shown in Figure 3.12, uses two opposed rectangles, with aligned centers. The aspect ratios are assigned as described above.

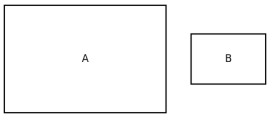


Figure 3.12.: Example stimulus for Task 8.

Task 9
Channel: (rectangular) area
Chart type: treemap
Task nine (Figure 3.13) introduces distractor rectangles in the form of a treemap. The marked rectangles are always the bottom-left and top-right ones and follow the same generation rules as in task eight. The random distractor treemap is generated using a recursive binary split treemap algorithm based on the work of Bederson et al. [10].

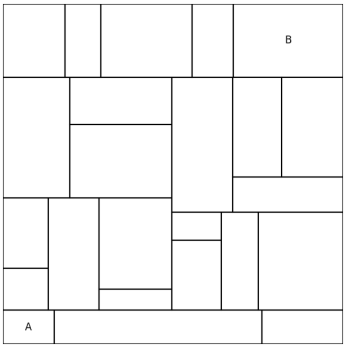


Figure 3.13.: Example stimulus for Task 9.

Task 10
Channel: (color) luminance
Chart type: squares
Task ten (Figure 3.14) uses two equally-sized squares with center-aligned positions, similar to task eight. The channel under evaluation is color luminance, where the s_i(3.1) values are used on a scale from 0 (white - max. luminance) to 100 (black - min. luminance).

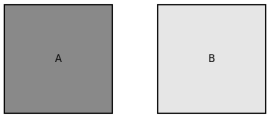


Figure 3.14.: Example stimulus for Task 10.

Task 11
Channel: (color) saturation
Chart type: squares
Task eleven (Figure 3.15) follows the same layout as task ten. The visual channel is changed to color saturation. This means the grayscale is replaced by a red saturation scale from 0 (white) to 100 (fully saturated red).

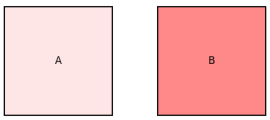


Figure 3.15.: Example stimulus for Task 11.

3.2.4. Stimuli Grouping and Cost-Aware Task Allocation

A significant methodological hurdle in evaluating modern MLLMs at scale is the financial and computational costs of Application Programming Interface (API) calls, particularly those involving image tokens and long reasoning chains. For this study, the Python-based image generation pipeline produces 45 distinct stimuli per task type. Each task type represents a specific visual encoding channel applied to various chart types.

If we were to evaluate every single model on all 45 images across six prompt variations, the token consumption would quickly exceed practical constraints, leading to prohibitive API costs and extended execution times. To maintain statistical validity within a reasonable real-world budget, we implemented a structured grouping strategy based on maximum variety in the underlying ratios.

This grouping is not arbitrary. The used data generation process relies on nine specific ratios between the two compared values in the stimuli (ranging from 0.18 to 0.83 given by the resulting pairs of s_i in equation 3.1).

The groups are constructed so that each group of nine images contains each of the nine ratio types exactly once. To achieve this, the 45 generated stimuli are first categorized into nine distinct buckets, corresponding to each of the nine proportional ratios. Because of the distribution in our stimuli generation, these buckets are of unequal sizes. The bucket for the smallest ratio contains only one element (The pair of the smallest and largest values), the second contains two and so on. Next, stimuli are drawn from each of the nine buckets to form five groups of nine images. Within each group, the stimuli are ordered sequentially from bucket one, the smallest and thus hardest ratio, down to bucket nine, the largest and easiest ratio to establish a clear progression of difficulty [1]. For buckets containing five or more stimuli, selections are made without replacement to maximize structural variety across groups. For smaller buckets with fewer than five stimuli, values are sampled with replacement and the requirement that each pair must be drawn at least once. The detailed grouping layout and the underlying values of the stimuli in each group are documented in

Table 3.1.: Overview of the six LLMs evaluated in this study. Properties include parameter count, model architecture (Dense vs. MoE), weight availability, context window size and perceived capability tier.

Model	Tier	Parameter Count	Architecture	Weights	Context Window
`gpt-5.4`	Frontier	Not specified	Dense	Closed	1M
`gemini-3.1-pro-preview`	Frontier	Not specified	MoE	Closed	1M
`kimi-k2.6`	Mid - Frontier	Not specified	MoE	Open	256K
`gemma-4-31b-it`	Mid	31B	Dense	Open	256K
`gemma-4-26b-a4b-it`	Small - Mid	26B/4B active	MoE	Open	256K
`nemotron-3-nano-omni-30b-a3b-reasoning (:free)`	Small	30B/3B active	Hybrid MoE	Open	256K

Appendix A.2.

During the evaluation phase, instead of running all 45 images through every prompt variation, we assign one of these stratified groups to each prompt. This reduces the number of inferences per model and task type from 270 (45 images $\times$ 6 prompts) to 54 (9 images $\times$ 6 prompts), an 80% reduction in API expenditure. Because each group of nine contains the full difficulty spectrum of ratios, the performance metrics derived from every prompt evaluation accurately reflect the model's capability across the entire difficulty range, maintaining statistical validity while respecting computational budgets.

3.3. Model Selection and Capability Tiers

To enable further analysis of model performance and capabilities, a range of models was selected based on several factors. The selection was made to provide a broad overview of current model capabilities, architectures and sizes, while keeping the thesis's scope manageable. The chosen models include frontier state-of-the-art models such as OpenAI's GPT-5.4 and Google's Gemini 3.1 Pro, which both feature an impressive one million token context window. While this Gemini version uses a MoE architecture, the GPT-5.4 version uses a dense transformer approach, enabling a comparison of architectures between the two top-tier models.

In this context, a dense transformer architecture means the model uses every neuron in its network to compute the output for each input token. Conversely, MoE models route tokens to specialized sub-networks, so only a small subset of neurons needs to activate for each token. MoE LLMs are often denoted as *Name_xb_ayb* (see gemma-4-26b-a4b-it in 3.1) where x denotes the total parameter count and y the active parameters.

```
Task: You are given a chart displaying quantitive
     attributes. Two elements in the chart are
     marked as ( A ) and ( B ). Your task is to {
     TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {
     APPROACH_DETAIL}.
Output Format: The output must be in this exact
     JSON format and cannot contain any other text:

   {
       "Is A smaller than B": true/false,
       "percentage": "XX%"
   }
Output Example:
   {
       "Is A smaller than B": true,
       "percentage": "50%"
   }
```

(a) Direct-Answer Prompt

```
Your role: You are an expert in chart
     interpretation, visual perception,
     information extraction and low-level chart
     QAs.
Task: You are given a chart displaying quantitive
     attributes. Two elements in the chart are
     marked as ( A ) and ( B ). Your task is to {
     TASK_DESCRIPTION}.
Approach: Think step by step about how you will
     identify the marked elements, estimate the
     quantitive values of them and calculate the
     percentage. Write your reasoning process down
      in the specified output field.
Output Format: The output must be in this exact
     JSON format and cannot contain any other text:

   {
       "Reasoning": "<Write your reasoning here>",
       "Is A smaller than B": true/false,
       "percentage": "XX%"
   }
Output Example:
   {
       "Reasoning": "I read the value of A. I read
            the value of B. I calculated the
           percentage. I gave the answer in the
           specified output format.",
       "Is A smaller than B": true,
       "percentage": "50%"
   }
```

(b) Reasoning and Role Prompt

Figure 3.16.: Modular prompt templates: baseline Direct-Answer prompt versus Reasoning and Expert-Role prompt.

As medium-sized models, Moonshot's Kimi 2.6 and Google's Gemma 4 31B were chosen. Kimi builds on MoE, whereas Gemma, in contrast to its larger Google rival Gemini, is a dense model. Both are fully open source and publicly available for local deployment.

The smallest model tier we evaluate features the 26B version of Gemma 4 and Nvidia's Nemotron 3 Nano Omni with 30B parameters. Interestingly, the smaller Gemma 4 model utilizes a MoE architecture, unlike its larger sister model. Nemotron uses a hybrid MoE variant. The lack of a dense model in the smallest of our tiers is worth pointing out, as the current landscape in this size range is dominated by MoE and hybrid models and they offer a better performance-to-compute ratio [21].

3.4. Prompting Design

It is well documented in the natural language processing literature that LLMs can exhibit markedly different performance profiles depending on the prompt instructions. In the context of graphical perception, we need to understand whether a model fails at a task because

its visual capabilities are limited or because the instructions did not elicit the model's full visual-reasoning potential.

To disentangle these factors, we designed a comprehensive suite of six different prompts. These prompts are categorically divided into three distinct conceptual axes: Expert Role prompts, CoT/reasoning prompts and duplication prompts [19, 20, 22]. Through our methodology, we aim to isolate visual capabilities from prompt sensitivity.

To effectively evaluate the MLLMs on the selected graphical perception tasks, the prompt design is a crucial factor. The six distinct prompt variations draw inspiration from the prompts proposed by Guo et al. [11] while mirroring the intent of the original questions in the foundational work by Cleveland and McGill [1] and Heer and Bostock [2]. It is worth noting that the exact phrasing of the questions used in these foundational studies is not fully documented in their respective papers. Therefore, the questions were reconstructed based on their methodology descriptions. Additionally, the raw prompt texts for all prompt variations and tasks are available in Appendix A.1.

The core prompt design remains simple and concise, avoiding overly complex context to minimize token cost and reduce potential confounding factors that could arise from verbose instructions. All prompts are constructed in a zero-shot manner for the core task instructions. This ensures that the evaluation measures the models' inherent, pre-trained graphical perception capabilities rather than their ability to perform in-context learning or pattern matching from provided examples. It is worth noting that this does not hold true for the output format template, which is added to all prompts twiceonce as an empty template and once as a filled-out example, to ensure the model generates the answer in the specified format. However, this one-shot part of the prompt does not affect the model's visual reasoning or estimation capabilities, but rather aims to ensure the output format is consistent and parsable. Since the model is not shown a visual stimulus with its correct answer, we still consider the overall prompt to be zero-shot.

The core structure of the prompts builds on a direct question format also used in the original human experiments. Specifically, they ask the model to identify the larger of two marked elements (A and B) and estimate the proportion between them as a percentage. Naturally, the exact phrasing of the question is adapted depending on the visual channel evaluated by the specific task. For example, while it makes sense to ask about the baseline or height for bar charts, the wording changes to area or angle for pie and bubble charts.

The prompt template is set up with a modular line-by-line structure to allow for easy removal or addition of single-line components. The separate elements are built using a $\langle Purpose \rangle : \langle Description \rangle$ syntax. This design choice was made to ensure the prompt components remain consistent across different variations. The components included in all prompt variations are the task instructions, the output format template and an filled-out answer response. Exemplary prompt configurations are shown in Figure 3.16, comparing the baseline Direct-Answer setup with the Reasoning and Expert-Role setup.

The variations of these prompts span three main categories: The models *approach* to solving the problem (Direct-Answer vs. Reasoning) [19], explicit instructions regarding the models *role* (No role vs. Expert role) [22] and whether the response is *duplicated* (No duplication vs.

Table 3.2.: Overview of the six prompt variations used in the experiments. Variations in approach (Direct vs. Reasoning), the use of an expert persona and prompt repetition (Duplication) are combined to generate six distinct prompts.

Prompt File / Name	Approach (Direct vs. Reasoning)	Expert Persona	Prompt Repetition (Duplication)
`No_Role_No_Reasoning.txt`	Direct-Answer	No Role	No Duplication
`Role_No_Reasoning.txt`	Direct-Answer	Expert-Role	No Duplication
`No_Role_Reasoning.txt`	Reasoning	No Role	No Duplication
`Role_Reasoning.txt`	Reasoning	Expert-Role	No Duplication
`Duplicate_No_Role.txt`	Direct-Answer	No Role	Duplication
`Duplicate_Role.txt`	Direct-Answer	Expert-Role	Duplication

Duplication) [20].

3.4.1. Direct-Answer vs. Reasoning

The foundational split in our prompt suite is the requested response generation strategy. The *Direct-Answer* condition is designed to provide an immediate, unreasoned response similar to the original question phrasing from Cleveland and McGill [1] and Heer and Bostock [2]. It asks the model only for the mark of the smaller element (A or B) and the numerical result, instructing it to output only the requested information without any supplementary text.

The *Reasoning Prompt* promotes explicit, structured reasoning before arriving at the final answer, leveraging the concept of CoT prompting [19]. This approach asks the model to articulate its visual analysis process step-by-step, such as identifying the baseline of a bar chart, locating the specific marks A and B, estimating their individual absolute heights and finally performing the division. While the reasoning prompt requires more tokens to generate and consequently incurs a higher cost, it aims to leverage the advanced reasoning capabilities of modern MLLMs. Previous research by Guo et al. [11] suggested that incorporating explicit reasoning steps can sometimes improve accuracy and more closely align model performance with human baselines on certain visual tasks. By testing these two distinct prompting approaches, we can analyze the models' sensitivity to reasoning explicitly imposed by the prompt.

3.4.2. Expert Prompting and Duplication

To generate the full suite of six prompts, these base strategies are augmented with role-playing and duplication techniques.

Role-playing prompts assign a specific *Expert-Persona* to the model, such as "an expert data analyst". This technique has been shown to help align the model's output toward more formal, precise and analytical language [22]. When a model adopts an analytical persona, its attention mechanism tends to place greater weight on tokens associated with logic and

precision, which we hypothesize may lead to more accurate numerical estimates when reading charts.

Duplication prompting involves deliberately repeating the prompt in its entirety within the same input, transforming the structure from $\langle Prompt \rangle$ to $\langle Prompt \rangle \langle Prompt \rangle$. The underlying mechanism, as detailed in recent research by Google researchers [20], is related to the causal nature of autoregressive transformers. Since these models process tokens from left to right, early tokens cannot attend to later instructions. Repeating the prompt allows tokens in the second instance to attend to the entire first instance, effectively giving the model a second pass to process the instructions and context. Empirically, the paper demonstrates that prompt repetition consistently improves performance on non-reasoning tasks (such as direct-answer Q&A or retrieval) across various model sizes. However, it offers little to no benefit when the model is prompted to use Chain-of-Thought or multi-step reasoning, as the generated reasoning steps already enable the model to internally attend to and reprocess the task instructions.

Grounded in these modern findings, six prompt variations are generated by combining the three strategies (Approach, Expert Persona and Prompt Repetition). We leave out the two potential variations that combine both Reasoning and Duplication, as the reasoning steps inherently involve internal question repetition and re-processing, which cancels out the benefits of explicit prompt duplication [20]. Table 3.2 lists the resulting six configurations.

We strategically allocate these six prompts based on model capability tiers. The duplication variants are exclusively applied to the free-tier and smaller open-weight models (Gemma 4 26B and Nemotron 3 Nano Omni), where the repeated reinforcement of task parameters is likely to yield the greatest improvements in output quality [20]. Conversely, the frontier models (GPT-5.4 and Gemini 3.1 Pro) are evaluated using the role-playing and reasoning prompts *without duplication*. This ensures that each model tier is tested without wasting computational resources on ineffective prompt structures.

3.5. Evaluation Metrics

The quantitative assessment of model performance relies on metrics identical to those used in the foundational human studies. This continuity is essential as without it, any comparison between human and AI perception would be fundamentally flawed. The primary objective of each experimental task is twofold. Binary classification (identifying which of the two marked elements is larger) is one aspect and continuous estimation (judging the exact percentage that the smaller element represents of the larger one) is the other.

For binary classification, we simply calculate accuracy as the percentage of trials in which the model correctly identifies the larger mark.

$$\begin{aligned} acc_t^m &= \frac{\sum_{i=1}^{n} I(y_i = \hat{y}_i)}{n} \\ t &\in 1, \ldots, 11 \\ m &\in \text{GPT-5.4}, \ldots, \text{Nemotron} \end{aligned} \tag{3.3}$$

This acts as a baseline sanity check. If an MLLM cannot reliably determine which bar in a chart is taller, its subsequent percentage estimations are practically meaningless.

However, the core of graphical perception theory is built on continuous estimation error. To quantify this, we could calculate the absolute error between the model's predicted percentage and the true mathematical percentage derived from the underlying generated data. Yet, absolute error alone is widely considered an insufficient metric for proportional perception tasks. The psychology of perception, as formalized by the Weber-Fechner law, dictates that humans process visual magnitudes logarithmically rather than linearly. This is also why the data generated by 3.1 are equally spaced on a log scale rather than a linear scale. An absolute error of 5% when estimating a 10% proportion represents a massive perceptual failure (a 50% relative deviation), whereas the exact same absolute error of 5% when estimating a 90% proportion is a minor, almost negligible inaccuracy.

To account for this non-linear reality, Cleveland and McGill [1] used the log-absolute error metric, which was subsequently adopted by Heer and Bostock [2]. We employ the exact same formula in our evaluation.

$$\text{log-absolute error} = \log_2\left(|\text{estimated percentage} - \text{true percentage}| + \frac{1}{8}\right) \tag{3.4}$$

The addition of the $1/8$ constant prevents the logarithm from evaluating to negative infinity in cases where the model perfectly guesses the exact percentage, effectively bounding the minimum possible error. By evaluating the MLLM using this log-absolute error, we not only measure how far off their estimates are in an absolute sense, but we scale those errors in a way that inherently respects the proportional nature of perceptual accuracy, placing heavier penalties on large deviations at small magnitudes and compressing the variance of minor inaccuracies.

3.5.1. Comparison Logic vs. Human Performance

A central, driving motivation of this thesis is to benchmark modern MLLM graphical perception directly against historical human results. Because we have recreated the stimuli generation rules, the visual constraints (such as distractor elements and spatial positioning) and the exact ground-truth ratio values used by Heer and Bostock [2], our dataset is a direct, modern expansion of theirs. This strict 1:1 structural alignment allows us to overlay our MLLM error distributions directly onto the collected human error distributions.

The comparison logic we employ is heavily reliant on non-parametric statistical methods. This is an intentional choice, recognizing that absolute error distributions in perception tasks are typically highly skewed and heavy-tailed, rather than conforming to a perfect Gaussian distribution. To compare the central tendencies of model performance against the human baseline, we rely on bootstrapped 95% confidence intervals for the mean log-absolute error. Bootstrapping involves resampling the error data with replacement thousands of times to empirically estimate the sampling distribution of the mean. This is a robust statistical technique that does not assume normality, making it well-suited to the asymmetrical nature of perceptual errors.

Furthermore, to determine whether the difference between a specific model's accuracy and human accuracy is statistically significant, we use the Mann-Whitney U test. Unlike a standard independent t-test, which compares means and assumes normality, the Mann-Whitney U test is a nonparametric test that assesses whether values in one sample are consistently ranked higher or lower than those in another sample. This makes it the ideal tool for comparing two independent groups with non-normal distributions. Through this rigorous, historically consistent statistical framework, we can state with confidence not just whether a model is "better" or "worse" than a human, but exactly how significant that perceptual gap is across different visual encodings, independent of statistical noise.

3.5.2. Establishment of a Channel Hierarchy

The ultimate synthesis of our experimental data culminates in the establishment of a visual encoding channel hierarchy specific to MLLMs. The human perceptual hierarchy forms the bedrock of modern data visualization design principles. The primary research question of this thesis asks a profound question. Do MLLMs inherently follow this same hierarchy?

To construct the MLLM channel hierarchy, we aggregate the mean log-absolute errors across all tasks mapped to a specific visual channel. For example, the error metrics from the pie chart tasks are aggregated to represent the angle channel. The tasks involving distinct squares or treemaps are aggregated to represent the area channel. Finally, the baseline-aligned bar chart tasks are aggregated to represent the position channel. We then sort these channels mathematically from the lowest mean log-absolute error (indicating the highest perceptual accuracy) to the highest (indicating the lowest accuracy).

This finalized ranking will be visualized using horizontal dot plots equipped with bootstrapped 95% confidence intervals, directly mirroring the visual presentation style popularized by Cleveland and McGill. By arranging the visual channels along a vertical axis and plotting the error magnitudes along the horizontal axis, the hierarchy becomes instantly legible and perfectly comparable to historical human data. We will generate individual hierarchies for each model, model tier as well as an aggregate overarching hierarchy across all tested models.

By comparing the resulting MLLM hierarchy with the established human hierarchy side by side, we can draw profound conclusions about the fundamental nature of machine vision. If the MLLM hierarchy matches the human hierarchy exactly, it would strongly suggest that the ViT architectures and the immense, human-generated datasets they are trained on have internalized the exact same geometric and spatial biases that evolutionary biology instilled in the human visual cortex. Conversely, if we discover significant inversions in the hierarchy (for instance, if models demonstrate a bizarre capability to evaluate area more accurately than angle, or length more accurately than position), it would imply that AI processes visual information in a fundamentally alien manner. Such findings would act as a catalyst for a massive paradigm shift in visualization design, suggesting that charts meticulously optimized for human consumption might actually hinder machine comprehension, thereby paving the way for a brand-new field of AI-centric, hybrid data visualization techniques.

3.6. Reproducibility and Usage

Ensuring the transparency, verifiability and repeatability of computational experiments is a fundamental cornerstone of modern computer science research, especially in an era characterized by rapidly updating, proprietary "black-box" AI models. The entire experimental setup underpinning this thesis, encompassing the Python-based matplotlib stimuli generation pipeline, the exact prompt template configurations and the automated API execution scripts, has been made fully publicly available under an open-source license. The complete repository, alongside execution instructions and raw JSON logs of model responses, can be accessed at `github.com/jakobzoebl/Bachelor_Thesis`. This comprehensive documentation is intended to empower the global research community to verify our findings, independently test newly released models against our established benchmarks and organically extend the methodology to novel visualization types or alternative AI architectures.

To guarantee that the exact stimulus images can be perfectly reconstructed without requiring the long-term storage of large archives of binary PNG files, the randomly generated distractor elements within the charts are strictly controlled via a seeded pseudo-random number generator. Throughout every single phase of this study, the global seed was set to 42. Any researcher executing our generation scripts with this specific seed will produce the identical set of 45 images per task type, the identical mapping of images into the five distinct stratified subsets and the identical allocation of mathematical ratios, ensuring flawless reproducibility of the core experimental conditions from the ground up. Furthermore, the exact prompt texts utilized during the evaluations are exhaustively listed and categorized in Appendix A.1 for immediate reference.

The entire experimental setup was constructed as a single Command Line Interface (CLI) tool for the purpose of having a single information flow pipeline. The tool lets the user choose which stimuli he wants to (re-)generate, which models he wants to evaluate, which tasks and prompts those models should be evaluated on and how the results should be aggregated. This gives the user full flexibility to customize the experiments and reproduce the results presented here. To get the same stimuli images, a default seed of 42 was used for the pseudo-random number generator.

The model connection was realized through the OpenRouter API, which provides a unified interface for accessing a wide range of LLMs. The entire code is open-source and available on GitHub.

4. Results

This chapter presents the experimental results of the graphical perception evaluation across six MLLMs. The findings answer the research questions in the order presented in Section 1.1. It additionally covers a supplementary spatial distance experiment in Experiment 2. It also covers cost-performance trade-offs and sensitivity to difficulty. A complete overview of all Log Absolute Error (LAE) results across models, tasks and prompt conditions is provided in the full results heatmap in Figure 4.1.

4.1. Dataset Overview

The primary experiment, combined with the distance manipulation task (Experiment 2), yielded 3,330 individual model responses. This spanned all six models, twelve task types and four to six prompting conditions per model. Of these responses, 3,297 were successfully parsed into valid numerical estimates. The 33 unparsed responses were excluded from all subsequent analyses. Responses with a percentage estimate exceeding 100% were treated as binary identification failures. Their LAE values were set to null and thus excluded from continuous LAE analyses but not from binary accuracy analyses.

A Shapiro-Wilk test was applied to the pooled LAE values. The data was aggregated across all models, tasks and prompts into a single unified dataset, yielding $n = 3{,}162$ valid observations. The resulting statistic is $W = 0.9669$. This is close to 1. The extremely large sample size yields a highly significant $p < 0.000001$. This confirms that even small deviations from normality are statistically detectable at this scale. This result justifies the exclusive use of non-parametric statistical methods. These methods include Kruskal-Wallis tests and Mann-Whitney U tests and Dunn's post-hoc comparisons and intraclass correlation and bootstrapped confidence intervals.

Table 4.1 presents the descriptive statistics of LAE per model, ordered by mean error. Note that this table and the channel hierarchy analysis in Section 4.2 explicitly exclude the Experiment 2 distance manipulation tasks ($n = 2{,}805$) to focus on the foundational visual encoding channels.

4.2. Aggregate Channel Hierarchy (RQ0)

Do MLLMs exhibit a generalized and consistent and measurable perceptual hierarchy across visual encoding channels for quantitative information?

Reason	Role	Model	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	Overall
x	x	Gemini 3.1 Pro	-0.59	-0.14	-0.49	-0.64	-0.32	1.15	-0.18	-1.29	0.26	5.70	5.70	0.83
		Gemma 27B	3.26	3.36	3.53	3.54	2.37	2.65	3.76	3.53	2.92	5.37	5.66	3.63
		Gemma 31B	3.61	1.92	4.96	2.95	2.50	2.47	2.87	2.63	2.85	5.70	5.70	3.47
		Kimi K2.6	2.01	2.63	2.10	2.73	2.12	2.33	2.02	2.28	2.93	5.23	5.56	2.90
		Nemotron	3.84	2.30	5.53	3.70	3.02	3.16	3.37	2.95	4.02	5.70	5.67	3.93
		GPT-5.4	2.68	3.16	4.68	3.07	3.84	2.69	3.48	2.82	2.60	5.42	5.21	3.60
		Prompt Overall	***2.31***	***2.16***	***3.28***	***2.52***	***2.06***	***2.41***	***2.55***	***2.13***	***2.59***	***5.52***	***5.58***	***3.01***
x	✓	Gemini 3.1 Pro	-0.55	-0.17	-0.65	-0.18	0.65	1.00	-0.46	-0.74	0.34	5.70	5.70	0.97
		Gemma 27B	3.55	3.34	3.54	4.01	2.75	3.34	2.97	3.36	3.15	5.60	5.46	3.73
		Gemma 31B	1.55	2.75	4.04	3.05	1.58	2.24	2.14	2.41	2.79	5.70	5.70	3.09
		Kimi K2.6	3.13	3.28	3.48	2.60	2.55	2.38	2.70	1.21	3.44	5.12	5.70	3.24
		Nemotron	4.10	1.87	4.88	3.87	2.74	2.44	3.87	3.12	4.59	5.70	5.70	3.90
		GPT-5.4	2.95	2.74	4.16	3.78	2.97	3.32	2.98	2.48	3.23	5.63	5.70	3.63
		Prompt Overall	***2.46***	***2.26***	***3.17***	***2.78***	***2.07***	***2.43***	***2.35***	***1.97***	***2.91***	***5.57***	***5.66***	***3.06***
✓	x	Gemini 3.1 Pro	-0.49	-0.73	-0.99	-0.93	-0.04	0.92	-1.00	-1.05	1.04	5.70	5.04	0.68
		Gemma 27B	2.41	2.15	2.87	2.50	2.64	3.73	2.88	3.99	3.31	5.70	5.42	3.42
		Gemma 31B	2.22	1.17	3.30	0.92	2.20	2.79	2.57	1.61	3.54	5.70	5.20	2.84
		Kimi K2.6	2.93	2.92	2.74	2.94	3.23	3.09	1.65	2.31	3.66	5.64	4.66	3.25
		Nemotron	4.14	2.49	4.88	3.69	3.90	3.79	2.63	2.89	3.89	5.51	5.70	3.96
		GPT-5.4	0.04	0.77	4.53	0.40	0.52	2.16	2.36	3.09	1.79	5.63	5.70	2.45
		Prompt Overall	***1.87***	***1.46***	***2.89***	***1.53***	***2.07***	***2.74***	***1.68***	***2.10***	***2.84***	***5.65***	***5.28***	***2.74***
✓	✓	Gemini 3.1 Pro	-1.15	-0.69	-0.01	-0.04	0.45	1.95	-0.84	-1.14	-0.17	5.70	5.70	0.89
		Gemma 27B	2.50	2.76	4.62	2.59	3.17	3.33	2.98	3.01	2.97	5.70	5.57	3.56
		Gemma 31B	2.64	1.57	3.67	1.71	2.08	2.84	2.95	2.01	3.52	5.70	5.70	3.13
		Kimi K2.6	2.31	2.95	2.02	2.70	2.01	2.48	2.62	1.79	3.88	5.56	5.61	3.08
		Nemotron	3.80	3.69	5.05	3.72	3.42	3.43	3.49	3.66	4.01	5.70	5.70	4.15
		GPT-5.4	-0.18	0.30	0.72	0.30	0.37	2.95	2.00	0.87	2.76	5.70	5.70	1.95
		Prompt Overall	***1.65***	***1.76***	***2.68***	***1.83***	***1.92***	***2.82***	***2.17***	***1.68***	***2.81***	***5.67***	***5.66***	***2.79***
Duplication	**Role**	**Model**												
✓	x	Gemma 27B	4.07	3.05	3.10	3.40	3.57	3.07	2.56	4.13	3.68	5.55	5.66	3.80
		Gemma 31B	3.45	2.70	3.81	2.24	2.01	3.61	2.49	3.37	2.26	5.69	5.70	3.39
		Nemotron	3.42	2.64	4.73	4.50	2.83	3.85	3.82	3.28	3.47	5.74	5.17	3.95
		Prompt Overall	***3.58***	***2.79***	***3.88***	***3.38***	***2.74***	***3.54***	***3.01***	***3.58***	***3.07***	***5.66***	***5.51***	***3.70***
✓	✓	Gemma 27B	3.83	2.68	4.18	3.26	2.38	3.04	3.58	3.20	2.98	5.35	5.39	3.62
		Gemma 31B	3.84	2.11	4.41	2.88	2.16	3.14	2.91	2.46	3.20	5.70	5.70	3.50
		Nemotron	3.99	2.50	4.69	3.52	3.50	2.47	3.88	3.05	3.66	5.61	5.44	3.85
		Prompt Overall	***3.90***	***2.41***	***4.47***	***3.23***	***2.75***	***2.86***	***3.45***	***2.89***	***3.30***	***5.55***	***5.51***	***3.67***
		Grand Total Overall	***2.35***	***2.04***	***3.22***	***2.38***	***2.17***	***2.72***	***2.41***	***2.22***	***2.86***	***5.60***	***5.54***	***3.05***

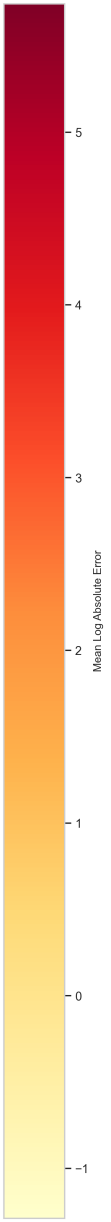
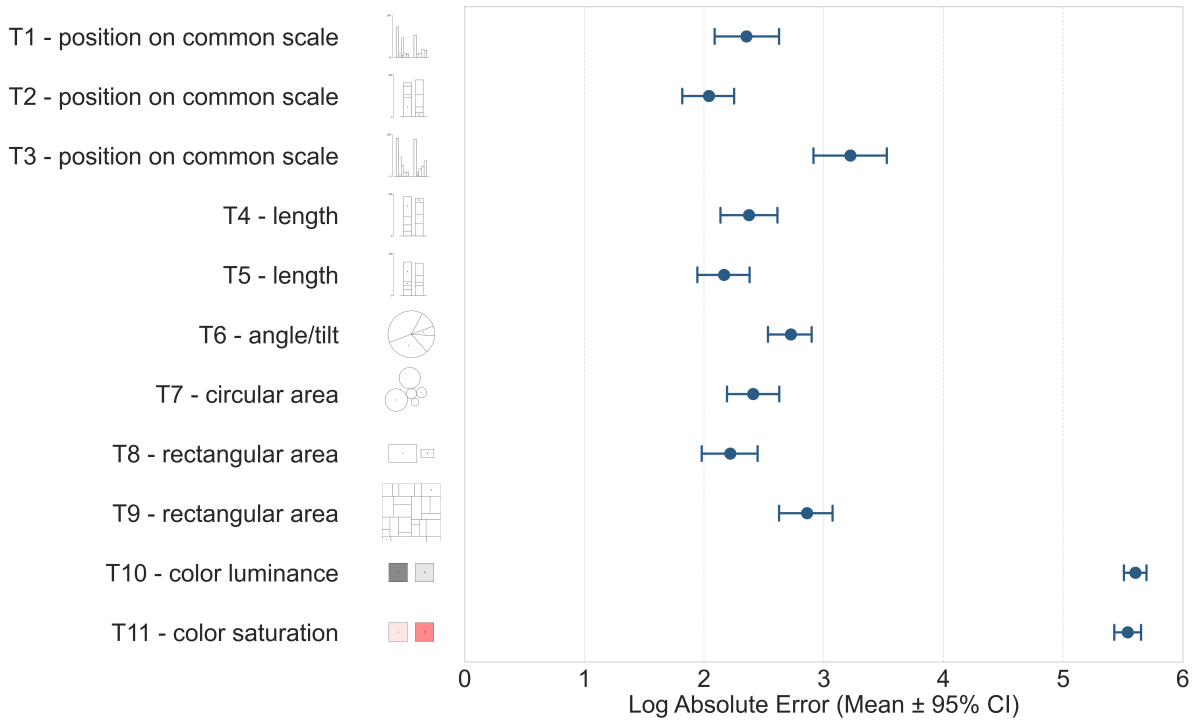
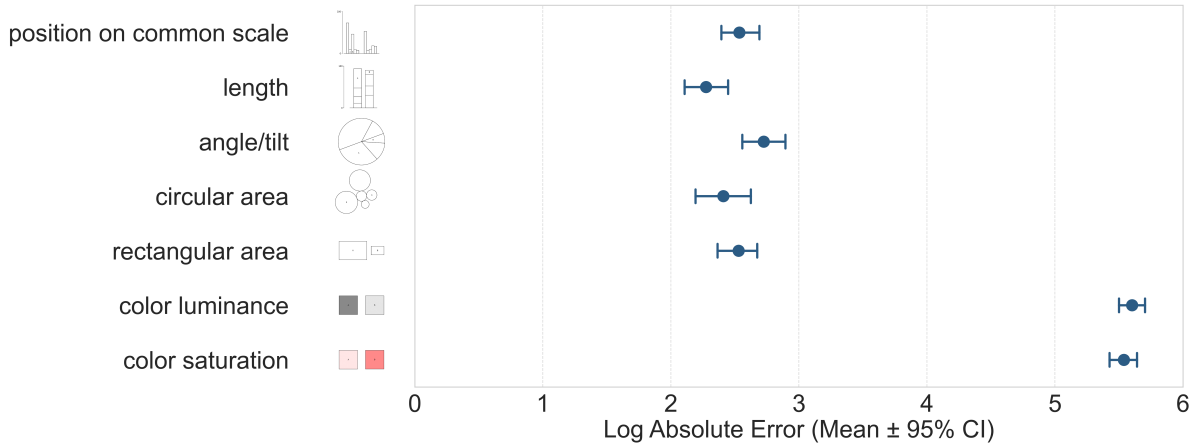


Figure 4.1.: Complete overview of mean LAE across all evaluated models, task types and prompt conditions. Lower values (lighter colors) indicate higher estimation accuracy.



(a) Per-Task LAE 95% CI



(b) Per-Channel LAE 95% CI

Figure 4.2.: Box-and-whisker plots showing the distribution of LAE per task type and per visual encoding channel in 95% CI, aggregated across all models and prompts. Lower values indicate higher estimation accuracy.

4.2.1. Task-Level and Channel-Level Error Distributions

The LAE distributions were analyzed at two levels of granularity. The first level is per individual task type as seen in Figure 4.2a. The second level is aggregated by the visual

Table 4.1.: Descriptive statistics of LAE per model, ordered by mean error (lower is better). N denotes the number of valid observations after exclusion of parse failures and invalid responses.

Model	N	**Mean LAE**	**Median LAE**	**SD**
Gemini 3.1 Pro	395	0.839	−0.007	2.592
GPT-5.4	368	2.886	3.137	2.238
Kimi K2.6	393	3.125	2.938	1.738
Gemma 4 31B	566	3.254	3.088	1.840
Gemma 4 26B	501	3.714	3.777	1.639
Nemotron	582	3.958	4.190	1.610

encoding channel as seen in Figure 4.2b. Extensive per-model breakdowns of estimation error across both task types and prompt conditions are provided in Appendix A.4 (Figures A.1–A.6). At the task level, the eleven task types produce clearly differentiated error profiles. Tasks 1 and 2 both test position on a common scale with adjacent or stacked bars. They yield comparatively low errors with $M = 2.354$ and $M = 2.040$, respectively. Task 3 also tests position on a common scale but with the two marked bars separated across two distinct bar groups. This produces a noticeably higher mean error with $M = 3.222$. This performance gap within the same visual channel is a key observation. This observation motivated Experiment 2 in Section 4.6.

A similar pattern emerges for the rectangular area channel. Task 8 presents two isolated rectangles. It yields consistently lower errors with $M = 2.218$ than Task 9. Task 9 embeds the marked rectangles within a treemap as distractor elements with $M = 2.860$. This suggests that the presence and complexity of distractor elements have a pronounced effect on model accuracy. This goes beyond the intrinsic difficulty of the visual channel used.

The length tasks of Tasks 4 and 5 show moderate and comparable error levels, with $M = 2.376$ and $M = 2.167$, respectively. The angle task in Task 6 and the circular area task in Task 7 fall within a similar range, with $M = 2.725$ and $M = 2.410$ respectively. Both color tasks produce dramatically higher errors. Task 10 for luminance has $M = 5.602$ and Task 11 for saturation has $M = 5.538$. They form a visually and statistically distinct group (see Figures 4.2a and 4.2b). This performance band is far removed from all spatial and geometric tasks.

4.2.2. The MLLM Channel Hierarchy

The task-level errors are aggregated to the channel level. This combines Tasks 1 to 3 for position and Tasks 4 to 5 for length. It also includes Task 6 for angle and Task 7 for circular area. It combines Tasks 8 to 9 for rectangular area. Finally, it includes Task 10 for luminance and Task 11 for saturation. The aggregate ranking is summarized in Table 4.2. Confidence intervals were computed via non-parametric bootstrapping with 10,000 iterations. Significance groups were derived from Dunn's post-hoc test with a Holm–Bonferroni correction. Channels sharing the same group letter do not differ significantly ($p > 0.05$). To isolate the impact of

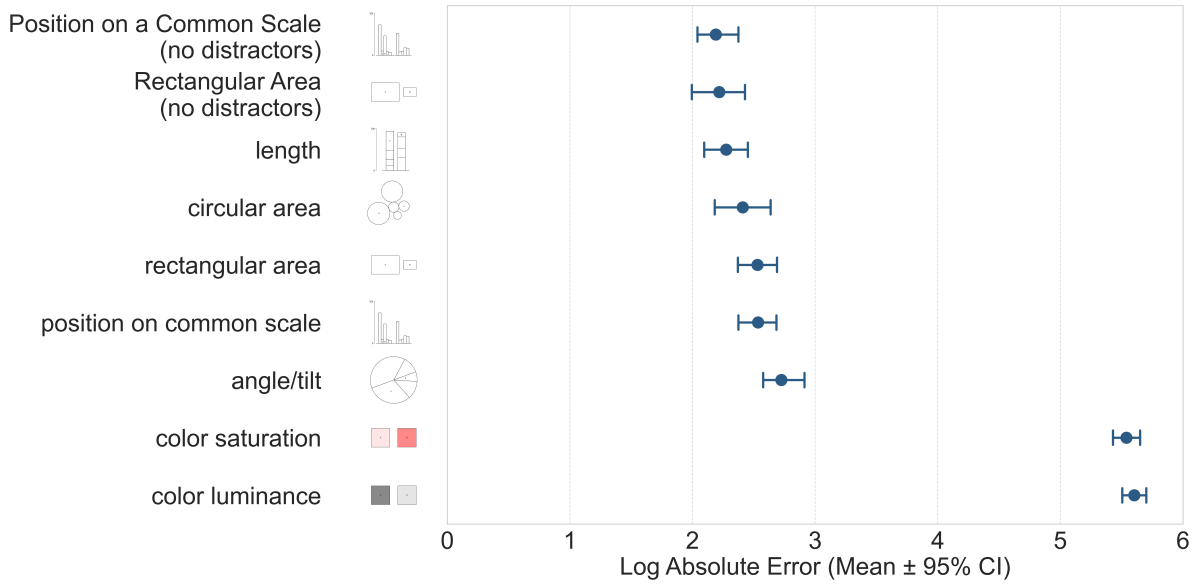


Figure 4.3.: Visual plot of the aggregate MLLM perceptual hierarchy. The layout identically mirrors Figure 4.2b but reorders the channels from lowest to highest error to reflect the actual performance hierarchy established by the models and explicitly includes the distractor-free variants.

Table 4.2.: Aggregate MLLM perceptual hierarchy ranked by mean LAE (lower = better). The distractor-free variants refer to Tasks 1 and 2 without Task 3 and Task 8 without Task 9.

Rank	Channel	Mean LAE	SD	*N*	95% Bootstrap CI	Sig.
1	Position on a Common Scale (no distractors)	2.190	1.989	502	[2.015, 2.364]	a
2	Rectangular Area (no distractors)	2.218	1.903	264	[1.991, 2.447]	a
3	Length	2.274	1.949	495	[2.101, 2.448]	a
4	Circular Area	2.410	1.811	253	[2.183, 2.635]	a
5	Rectangular Area	2.529	1.851	513	[2.368, 2.690]	a
6	Position on a Common Scale	2.534	2.181	753	[2.376, 2.692]	a
7	Angle/Tilt	2.725	1.402	253	[2.554, 2.895]	a
8	Color Saturation	5.539	0.916	269	[5.427, 5.645]	b
9	Color Luminance	5.602	0.827	269	[5.502, 5.696]	b

distractor elements, the tabular breakdown and the visual plot also include distractor-free variants. This is done for channels that contain both distractor and non-distractor task types. The visual hierarchy is plotted in Figure 4.3. Notably, this visualization identically mirrors the layout of Figure 4.2b but reorders the channels from lowest to highest error to match the

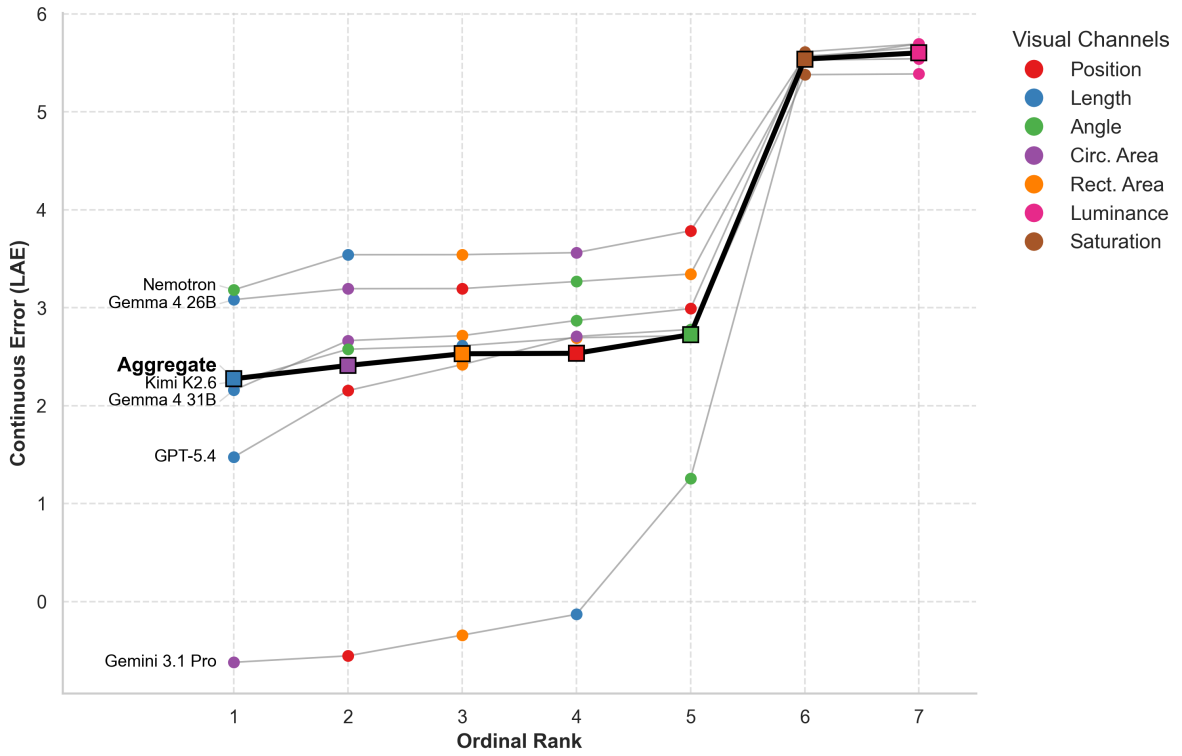


Figure 4.4.: Hockey stick plot showing the relationship between ordinal rank and continuous LAE for each model and the aggregate. The steep elbow between Rank 5 and Rank 6 reveals that the ordinal hierarchy compresses a large performance gap into a single rank step.

models' performance hierarchy and explicitly adds the distractor-free channel variants. The aforementioned Figures 4.2a and 4.2b are strictly ordered according to the human hierarchy.

A Kruskal-Wallis omnibus test confirms a highly significant main effect of the visual encoding channel on estimation error. The results are $H(6) = 1006.53$ and $p < 0.0001$ with an effect size of $\epsilon^2 = 0.3590$. This indicates that approximately 35.9% of the variance in LAE is attributable to the visual channel used. This represents a substantial effect.

4.2.3. Pairwise Channel Comparisons

A post-hoc Dunn's test with Holm-Bonferroni correction was conducted. This was done to identify which channel pairs differ significantly. The results reveal a clear bipartite structure.

- **Color vs. geometric channels** Both color channels perform significantly worse than every geometric channel with $p < 0.0001$ for all pairwise comparisons. The gap between the best color channel in saturation with $M = 5.539$ and the worst geometric channel in angle with $M = 2.725$ spans nearly 2.8 LAE units.

- **Within geometric channels** All pairwise differences among the five geometric channels are statistically non-significant with $p > 0.05$. Their bootstrapped confidence intervals overlap substantially as seen in figure 4.3.

This pattern indicates that MLLMs exhibit a coarse bipartite hierarchy rather than a fine-grained linear ranking. They process spatial and geometric encodings with comparable accuracy. They show a severe deficit in decoding continuous color information. This structural bipartite division is visually evident in Figure 4.4, which shows a sharp elbow between geometric and color channels for every single model as well as the aggregated line.

4.2.4. Cross-Model Concordance

We wanted to assess whether the observed hierarchy is shared across models or driven by a single dominant model. We computed the Intraclass Correlation Coefficient (ICC). The ICC operates on the actual continuous LAE values, unlike rank-based concordance measures such as Kendall's W. This is important because the geometric channels are separated by very small absolute differences. The full range spans only 0.45 LAE units. This means that minor fluctuations can flip ordinal ranks without reflecting meaningful performance differences.

The ICC across the six models' mean LAE profiles for the seven channels is ICC $= 0.822$ with $F(6, 30) = 28.66$ and $p < 0.0001$. This indicates strong and statistically significant agreement on the actual error magnitudes. This confirms that the bipartite hierarchy is not an artifact of any single model. It reflects a shared pattern across architecturally diverse MLLMs, as demonstrated by the consistently shaped curves in Figure 4.4.

The exact orderings within the geometric cluster vary. This reflects the non-significant pairwise differences. All six models consistently place both color channels at the bottom. The individual model rankings can be inferred from Figure 4.3, which already presents the channels in their aggregate performance order.

4.3. Comparison to Human Benchmark (RQ2)

How does MLLM low-level graphical perception accuracy compare to human performance both in terms of absolute estimation error and the derived ordering of visual encoding channels?

We compared MLLM performance against humans by evaluating the aggregate model channel hierarchy against the established human perceptual hierarchy from Cleveland and McGill [1] and Heer and Bostock [2] (shown in Figure 4.5). The aggregate MLLM hierarchy is presented in ranked order in Figure 4.3.

A rank correlation analysis was performed to quantify the alignment between the human and aggregate MLLM hierarchies. The results show a moderate but statistically nonsignificant correlation. We calculated Kendall's $\tau = 0.4286$ with $p = 0.2389$ and Spearman's $\rho = 0.5714$ with $p = 0.1802$.

The lack of statistical significance indicates that the MLLM hierarchy does not replicate the human hierarchy. Both hierarchies agree at the extremes. They place spatial channels above color channels. The internal ordering diverges substantially.

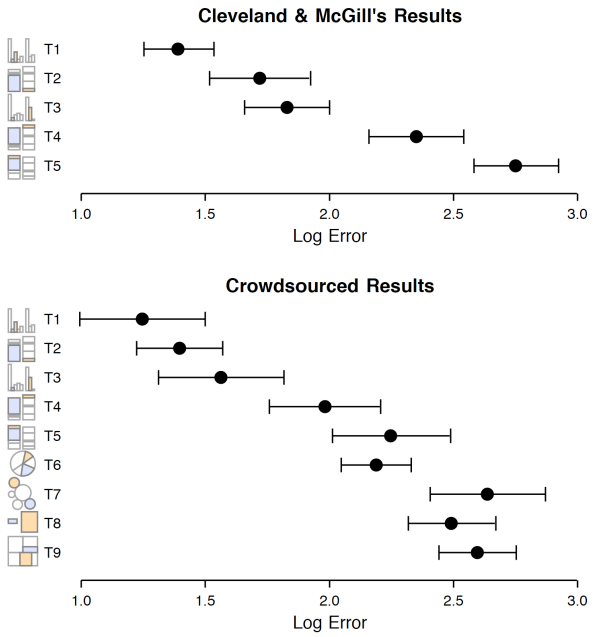


Figure 4.5.: Historical human graphical perception benchmark results from Cleveland and McGill [1] (top) and the crowdsourced replication by Heer and Bostock [2] (bottom).

Two key rank inversions drive the misalignment. First *position on a common scale* is the highest-ranked channel in human perception. It drops to rank 4 in the MLLM hierarchy. Second, *area-based channels* are ranked substantially higher by MLLMs than by humans. The differences between the geometric channels are not statistically significant in the MLLM data. This was discussed in Section 4.2. The exact rank positions within the geometric cluster should therefore be interpreted with caution. More broadly, these ordinal rankings are a limited proxy for the actual continuous differences in performance. A deeper examination of this distortion is provided in Section 5.2.

4.4. Influence of Prompt Design (RQ1)

To what extent does prompt design influence MLLM performance on low-level graphical perception tasks and alter the relative ranking of visual encoding channels?

4.4.1. Decomposed Prompt Axis Effects

The influence of prompt design was evaluated along three independent axes using Mann-Whitney U tests. Each test evaluates a single binary factor while controlling for the others. The results in Figure 4.6 reveal a clear distinction between cognitive and superficial prompting modifications.

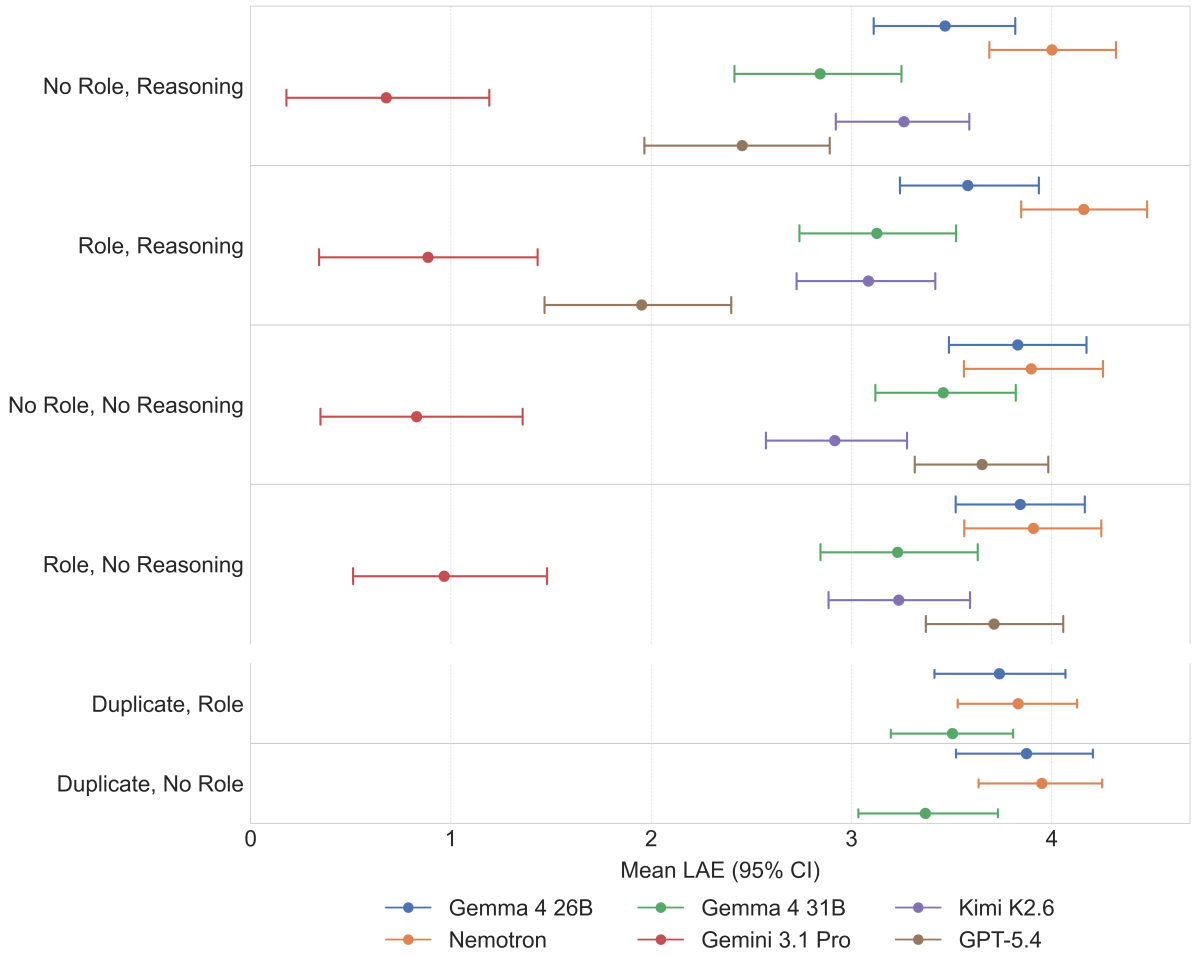


Figure 4.6.: LAE distributions across prompting conditions. The effect of prompt design on estimation error across all models and task types.

Reasoning vs. Direct Prompting Prompts instructing models to generate step-by-step reasoning produced lower estimation errors overall. As shown in the aggregate performance results in Figure 4.1, reasoning prompts without a persona role reduced the mean LAE from 3.01 (direct) to 2.74 (reasoning). Similarly, when an expert persona was assigned, reasoning prompts reduced the mean LAE from 3.06 to 2.79. As detailed in the token usage heatmap (Figure A.8 in Appendix A.5), the reasoning condition successfully induced deeper processing, generating approximately 1.28× as many output tokens on average as the direct-answer condition. This increased token generation corresponds with an improvement in estimation accuracy: the reasoning condition with $N = 1{,}344$ and $M = 2.600$ outperformed the direct-answer condition with $N = 1{,}818$ and $M = 3.123$. The statistical results show $U = 1{,}072{,}876.0$ and $p < 0.000001$ with a Cliff's delta of $\delta = -0.1218$. The negative delta confirms a downward shift in the error distribution. The overall effect is statistically significant. It is not uniform across models. GPT-5.4 with $\Delta = 1.478$ and both Gemma models with $\Delta = 0.296$ and

0.411 profit substantially from reasoning prompts. Gemini 3.1 Pro remains indifferent with $\Delta = 0.116$. Nemotron performs worse when instructed to reason with $\Delta = -0.184$ (see individual per-prompt error distributions in Appendix A.4, Figures A.1(b)–A.6(b)).

Expert-Persona vs. No Persona Assigning an expert persona, like an expert data analyst, to the model had no measurable effect. The role condition with $N = 1{,}578$ and $M = 2.909$ and the no-role condition with $N = 1{,}584$ and $M = 2.892$ are statistically indistinguishable. The statistical results show $U = 1{,}256{,}504.5$ and $p = 0.7932$ with Cliff's $\delta = 0.0054$. The near-zero effect size indicates complete overlap between the distributions.

System Prompt Duplication vs. No Duplication Repeating the prompt instructions within the system message was tested exclusively on the smaller models Gemma 4 26B and Nemotron. It did not significantly affect performance. The duplication condition with $N = 548$ and $M = 3.708$ and the no-duplication condition with $N = 617$ and $M = 3.563$ yielded $U = 177{,}128.5$ and $p = 0.1591$ with Cliff's $\delta = 0.0477$. Prompt reinforcement through repetition does not improve visual estimation accuracy.

4.4.2. Hierarchy Robustness Across Prompting Conditions

A critical question is whether different prompting strategies alter the relative ordering of the visual channels. We computed the Intraclass Correlation Coefficient (ICC) across the mean channel errors derived under each prompting condition to test this. The agreement is nearly perfect with ICC $= 0.9500$ and $F(6, 30) = 114.93$ and $p < 0.000001$.

This confirms that the underlying channel hierarchy established in Section 4.2 is completely robust to prompt variations. Figure 4.7 illustrates this stability visually. CoT reasoning shifts the absolute error curve downward for some models. It does not rearrange the relative difficulty ordering of the visual channels. Prompt engineering acts as a decoder calibration mechanism. It does not alter the models' intrinsic visual encoding capabilities.

4.5. Impact of Model Scale and Architecture (RQ3)

To what extent does the capability tier or architecture of an MLLM affect its graphical perception performance?

4.5.1. Individual Model Performance

Table 4.3 presents the individual model rankings based on mean LAE. The table includes post-hoc pairwise significance groupings derived from Dunn's test with Holm correction.

Gemini 3.1 Pro is a clear outlier. It achieves a mean LAE of 0.840. This is more than two full units below GPT-5.4. Its median LAE is -0.007. This indicates that most of its estimates fall within a very small error margin. This performance gap is statistically significant relative to every other model, with $p < 0.001$. The remaining five models form two overlapping clusters.

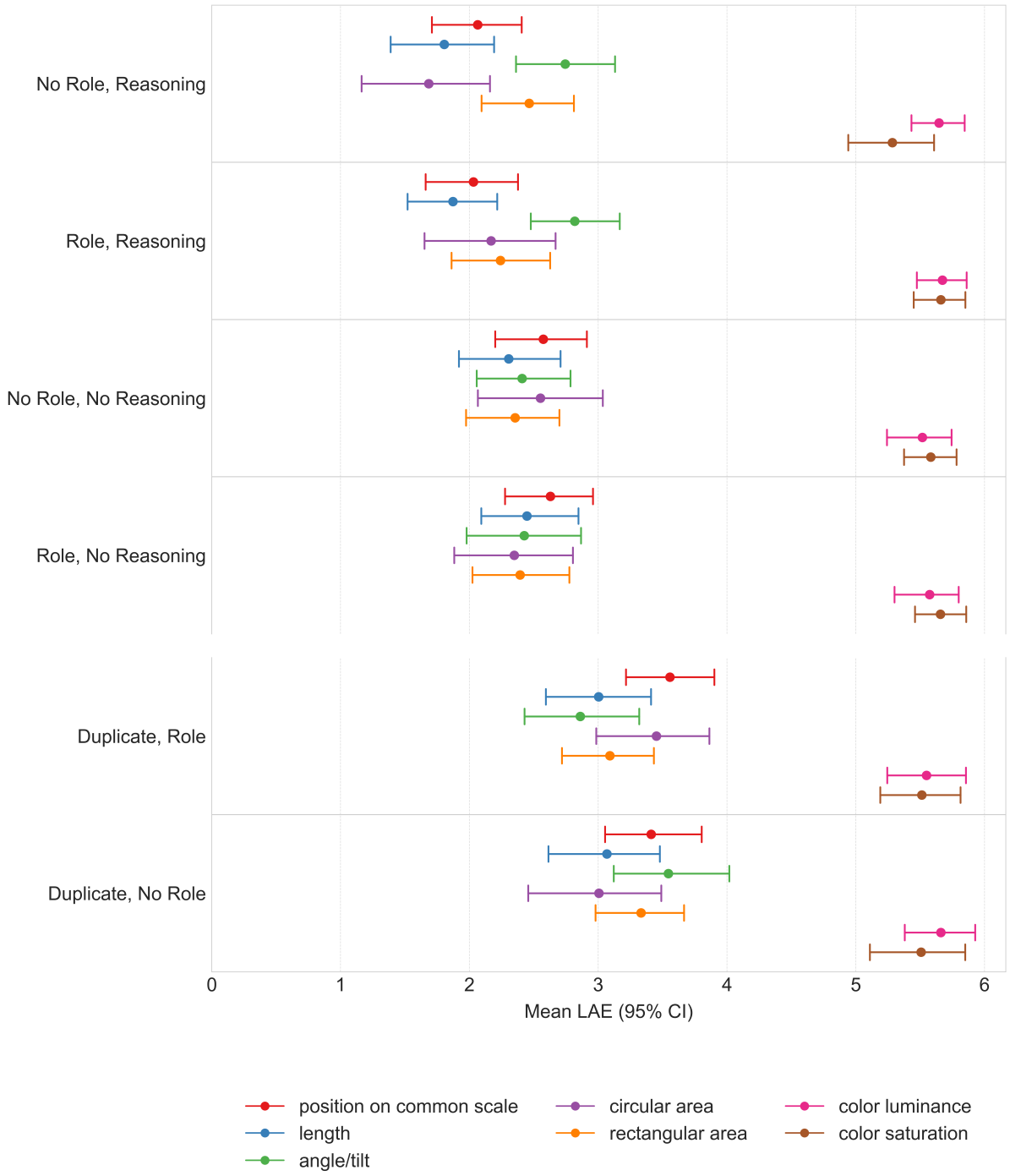


Figure 4.7.: The robustness of the visual encoding hierarchy across different prompting conditions at the channel level. Each prompt variant maintains the same structural ordering.

Rank	Model	Mean LAE	Pairwise Significance
1	Gemini 3.1 Pro	0.840	Differs significantly from all other models with $p < 0.001$
2	GPT-5.4	2.886	No sig. diff. from Kimi/Gemma 31B. Outperforms Gemma 26B and Nemotron
3	Kimi K2.6	3.125	No sig. diff. from GPT-5.4/Gemma 31B. Outperforms Gemma 26B and Nemotron
4	Gemma 4 31B	3.254	No sig. diff. from GPT-5.4/Kimi. Outperforms Gemma 26B and Nemotron
5	Gemma 4 26B	3.714	No sig. diff. from Nemotron. Outperformed by all others
6	Nemotron	3.958	No sig. diff. from Gemma 26B. Outperformed by all others

Table 4.3.: Model performance ranking by mean LAE with post-hoc pairwise significance groupings from Dunn's test with Holm correction. Lower LAE indicates better performance.

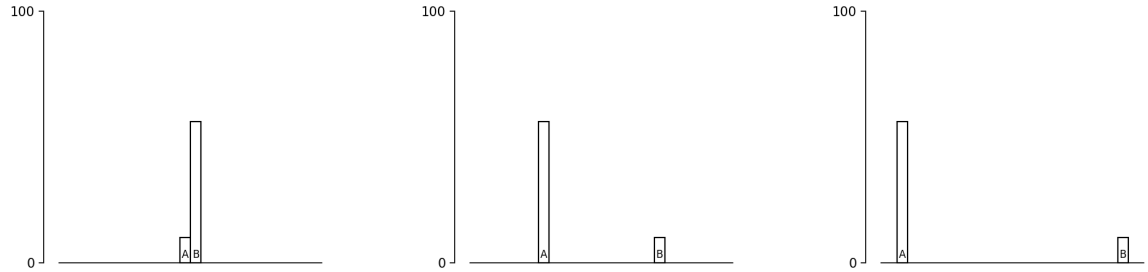
The first is a mid-accuracy group containing GPT-5.4, Kimi K2.6 and Gemma 31B. The second is a lower-accuracy group containing Gemma 26B and Nemotron. Detailed task-level and prompt-level error distributions for each individual model are presented in Appendix A.4 (Figures A.1–A.6).

4.5.2. Capability Tier Analysis

The six models were grouped into three capability tiers based on parameter count and context window size and developer positioning. These are *Frontier* with Gemini 3.1 Pro and GPT-5.4. Then *Mid* with Kimi K2.6. Finally *Small* with Gemma 4 31B and Gemma 4 26B and Nemotron. These groupings reflect the models' relative positioning in the landscape.

A Kruskal-Wallis omnibus test reveals a highly significant effect of capability tier on estimation error with $H(2) = 272.23$ and $p < 0.0001$. Pairwise Mann-Whitney U comparisons confirm that each tier significantly outperforms the one below it. The effect sizes indicate a clear gradient. The frontier-to-small gap is $\delta = -0.4055$. This is more than twice the mid-to-small gap of $\delta = -0.1742$. This suggests diminishing returns in graphical perception accuracy as capability decreases.

Gemini 3.1 Pro significantly outperforms GPT-5.4 within the Frontier tier. Gemma 4 31B overperforms for its size within the Small tier. It matches the performance of the Mid-tier Kimi K2.6 and approaches GPT-5.4.



(a) Distance Level D1 (Gap 0) (b) Distance Level D3 (Gap 10) (c) Distance Level D5 (Gap 20)

Figure 4.8.: Example stimuli from Experiment 2 demonstrating the variation in spatial separation between the two target bars across distance levels D1 (no gap), D3 (medium gap) and D5 (maximum gap).

4.5.3. Architecture Comparison

Evaluating the impact of model architecture requires caution. The performance difference between architectural approaches is heavily confounded by the models' capability tiers. Gemini 3.1 Pro stands out as an outlier in the dataset. A clean aggregation of architectural performance across the entire dataset is not possible. A direct statistical comparison between MoE and Dense architectures yields misleading results. These results would be driven entirely by Gemini.

We can utilize specific model pairs as proxies instead. The two Gemma models serve as proxies for smaller models with different architectural refinements. Gemini 3.1 Pro and GPT-5.4 serve as proxies for frontier models. The results indicate that the capability scale currently outweighs structural architectural differences in determining the accuracy of graphical perception.

4.6. Spatial Distance Effects (Experiment 2)

Section 4.2 noted a within-channel performance split for position and rectangular area. Tasks with adjacent or isolated elements yielded lower errors than tasks with spatial separation or distractors. In Experiment 2, we conducted a follow-up experiment to test whether the spatial distance between elements explains this variation.

Experiment 2 uses a simplified bar chart with only two bars. It has no distractor elements. It varies only the spatial separation between the bars across five distance levels (see Figure 4.8). These range from D1 (closest) to D5 (farthest). All six models were evaluated on this task using the same prompting conditions as the main experiment. This yielded 357 valid observations.

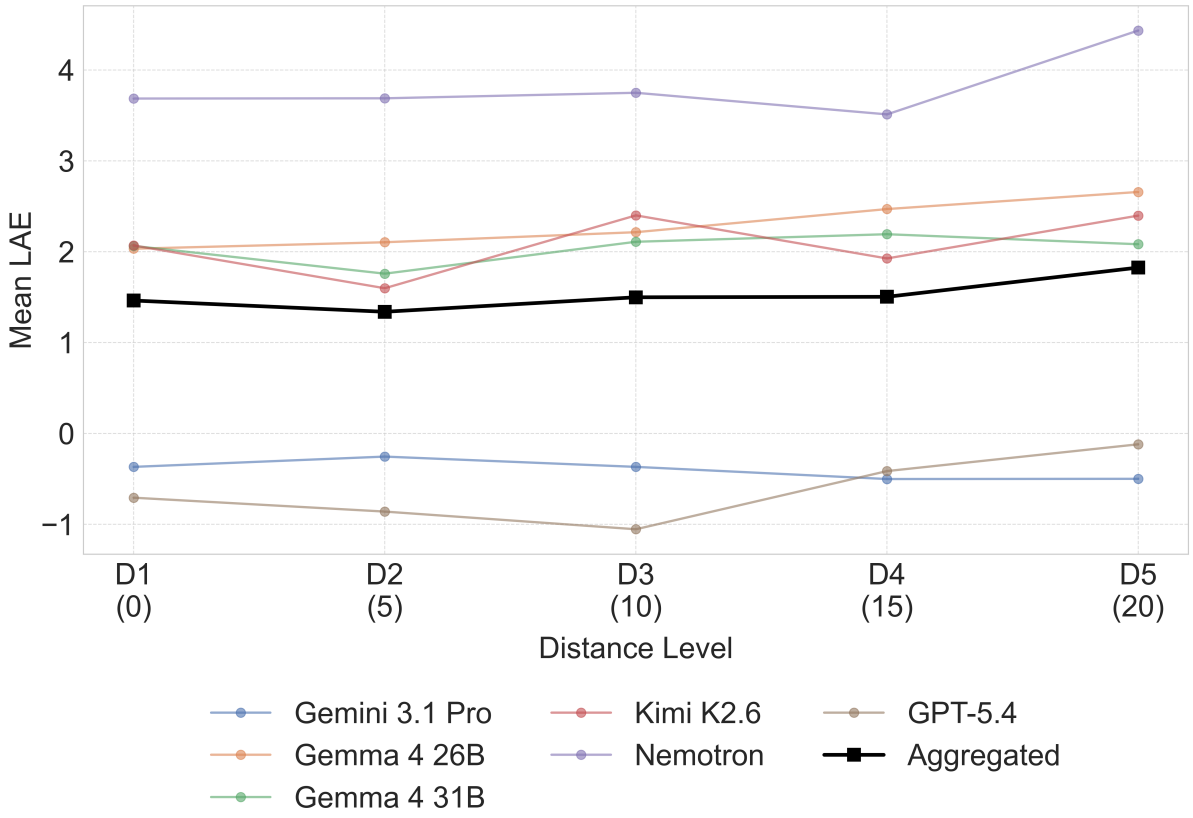


Figure 4.9.: Experiment 2 results. The effect of spatial bar distance on continuous estimation error.

4.6.1. Effect on Continuous Estimation Error

Figure 4.9 presents the relationship between spatial distance and continuous estimation error.

Spatial distance has no significant effect on the precision of continuous estimation, with $H(4) = 3.295$ and $p = 0.510$. The mean LAE values across the five distance levels are closely clustered with no systematic trend. There appears to be a slight trend toward worse performance across most models. However, Gemini managed to improve on larger distances.

4.6.2. Effect on Binary Accuracy

Distance does not affect the precision of continuous estimation. A Chi-Square test of independence reveals a significant association between distance and binary accuracy with $\chi^2(4) = 11.02$ and $p = 0.026$. Models achieve 100% binary accuracy at closer distances. They show degraded performance at larger separations (see Figure A.9 in Appendix A.6).

4.6.3. Implications for the Within-Channel Performance Gap

The absence of a distance effect on continuous error strongly suggests that spatial distance alone is not responsible for the performance gaps observed in the main experiment. The presence of distractor elements drives the elevated errors instead. Tasks 1, 2 and 8 consistently show better performance because they lack distractors. Tasks 3 and 9 show reduced overall performance due to the introduction of distractors. This confirms that distractors reduce the overall performance of MLLMs. Spatial distance does not.

4.7. Supplementary Analyses

4.7.1. Cost-Performance Trade-Off

We analyzed the relationship between API cost and estimation accuracy. This was done to evaluate the practical efficiency of deploying MLLMs for graphical perception tasks. A Spearman rank correlation between each model's mean API cost per query and its mean LAE reveals a near-perfect negative correlation with $\rho = -0.9429$ and $p = 0.005$. Better graphical perception is strongly associated with higher operational costs. The detailed heatmaps for cost and token usage are provided in Appendix A.5 in Figures A.7 and A.8.

The cost per correct binary identification varies by over four orders of magnitude across the evaluated models. It ranges from no cost for the free tier of Nemotron to €0.054948 for Gemini 3.1 Pro. This trade-off underscores a practical constraint for deployment. Frontier models deliver substantially higher estimation accuracy. Their cost profile may be prohibitive for large-scale automated workflows.

4.7.2. Difficulty Sensitivity and Weber's Law

Human graphical perception is known to follow Weber's Law. Estimation error increases predictably as the true ratio between compared elements approaches extreme proportions [1]. We computed Spearman rank correlations between the true percentage and the LAE for each model individually and across the full dataset to test whether MLLMs exhibit a comparable sensitivity.

Figure 4.10 presents the relationship between difficulty (true percentage ratio) and estimation error per model (the per-prompt breakdown is provided in Figure A.10 in Appendix A.7).

The overall correlation across all models is negligible and non-significant with $\rho = -0.013$ and $p = 0.495$. Only two models show statistically significant sensitivity at the individual model level.

- **Gemini 3.1 Pro** $\rho = 0.136$ and $p = 0.007$. This is a small but significant positive correlation. It means that the error increases as the true proportion becomes more extreme. This is the only model exhibiting a pattern reminiscent of human Weber-like sensitivity.
- **Gemma 4 26B** $\rho = -0.109$ and $p = 0.015$. This is a weak negative correlation in the opposite direction.

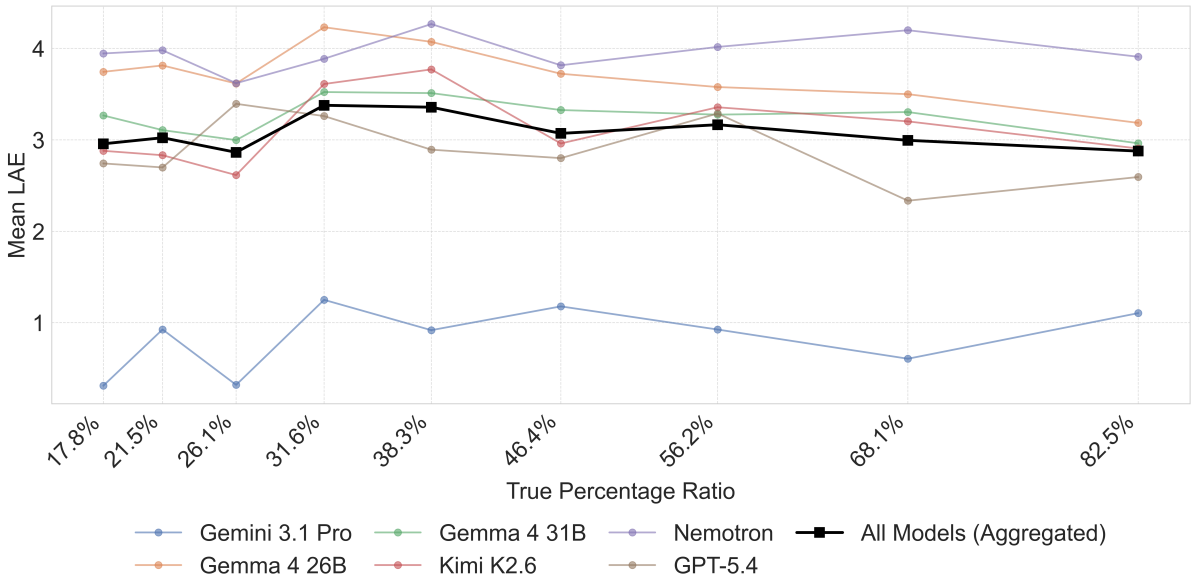


Figure 4.10.: Relationship between task difficulty (true percentage ratio) and LAE per model.

All other models show no significant sensitivity to the true ratio. This overall lack of difficulty sensitivity contrasts with the well-documented pattern in human perception. Estimation variance increases predictably with more extreme ratios in humans [1, 2].

4.7.3. Token Usage

We analyzed the relationship between token consumption and estimation error. This was done to verify that reasoning prompts improve accuracy through deeper processing rather than simply generating more output tokens. The Spearman correlation between each model's mean completion tokens and its mean LAE is negligible with $\rho = -0.086$ and $p = 0.872$. As shown in the token usage heatmap (Figure A.8 in Appendix A.5), reasoning prompts without a persona role generate a mean of 3218.7 completion tokens compared to 2500.3 for direct-answer prompts without a role (an overhead factor of 1.29×). When a persona role is introduced, reasoning prompts generate a mean of 2676.6 completion tokens compared to 2109.2 for direct-answer prompts with a role (an overhead factor of 1.27×). This indicates that the accuracy improvement from reasoning in Section 4.4 comes at a moderate but consistent token cost across role conditions.

5. Discussion

The results of this study could suggest that MLLMs possess substantial graphical perception capabilities. Their perceptual behavior might diverge from human heuristics. This chapter contextualizes these findings within the broader literature on machine vision. It discusses their potential implications for automated chart understanding and visualization design.

5.1. The Bipartite Hierarchy and Implications for Visualization Standards

A central finding of this work suggests a possible misalignment between the aggregate MLLM hierarchy and the established human benchmark in RQ2. Humans exhibit a fine-grained linear ranking of visual channels with *position on a common scale* typically performing best [1]. The evaluated MLLMs exhibit a coarse bipartite division instead. They appear to process all geometric encodings with statistically comparable accuracy. They seem to struggle significantly with continuous color encodings in RQ0.

This divergence could suggest that visualizations intended for automated processing or agentic retrieval might need to be optimized differently than human-facing charts. The relative weakness of MLLMs on position tasks at Rank 4 and their comparable strength on area tasks at Ranks 2 and 3 indicates that traditional human guidelines might not strictly apply. Human guidelines often prefer bar charts over area charts. Since MLLMs might process area and length with comparable fidelity, we could use treemaps or bubble charts more freely for AI consumption. This finding could impact visualization standards by encouraging a dual-design approach. Data graphics designed for LLMs could leverage a wider array of geometric channels without the performance penalty observed in humans. The results also suggest that continuous quantitative ratios from color saturation or luminance should be avoided in LLM-facing data graphics.

5.2. Distortion of Performance Through Ordinal Channel Rankings

The channel hierarchy established in Section 4.2 assigns each visual encoding channel an ordinal rank from 1 to 7. While this ranking provides a convenient summary, it compresses the underlying continuous differences in performance in a way that can be misleading. Within the geometric cluster, the five channels spanning Ranks 1 through 5 are separated by a total of only $\sim$0.45 LAE units. The gap between the worst geometric channel and the best color channel, however, spans nearly 2.8 LAE units. An ordinal ranking treats these as equally

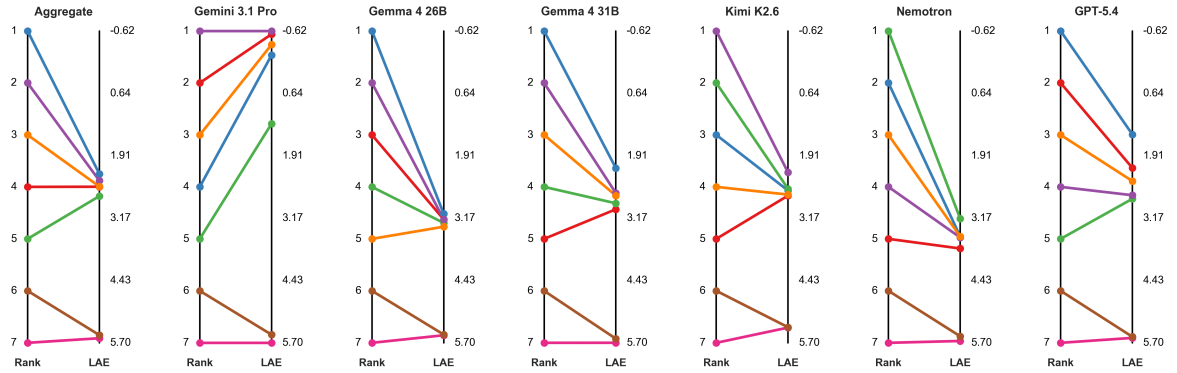


Figure 5.1.: Slopegraph comparing ordinal rank (left axis) to continuous LAE (right axis) for each model and the aggregate. Lines that cross or converge between the two axes indicate channels whose ordinal rank does not reflect their actual performance distance.

spaced steps, obscuring the fact that the performance difference between Rank 5 and Rank 6 is roughly six times larger than the entire spread across Ranks 1 through 5.

As introduced previously, Figure 4.4 visualizes this distortion directly. Each line traces a single model's channel performance from Rank 1 to Rank 7, plotting the ordinal rank on the horizontal axis against the continuous LAE on the vertical axis. If the ordinal ranking faithfully represented the actual performance differences, the lines would rise roughly linearly. Instead, every model exhibits a pronounced hockey stick shape. The geometric channels cluster tightly in the lower-left region with near-flat slopes and then a steep elbow appears between Rank 5 and Rank 6 as the transition to the color channels produces a dramatic jump in error. The bold aggregate line confirms that this pattern is not model-specific but structural.

The slopegraph in Figure 5.1 reinforces this observation from a complementary angle. For each model, the left vertical axis displays the ordinal rank and the right vertical axis displays the continuous LAE. Lines connecting matching channels across the two axes reveal how the transition from rank to continuous scale reshuffles the visual spacing. Across all models, the geometric and color-based channels consistently separate into two distinct performance clusters along the continuous axis, even when their ordinal ranks suggest a more gradual progression. The slopegraph makes it visually evident that the bipartite grouping, geometric channels versus color channels, is the dominant structure in the data rather than any fine-grained linear ordering.

These visualizations jointly caution against over-interpreting the precise ordinal positions within the geometric cluster. A model ranking circular area at Rank 2 and position at Rank 4 does not imply a meaningful performance advantage of area over position. The continuous error values for these channels overlap substantially and their pairwise differences are statistically non-significant as established in Section 4.2. The ordinal hierarchy remains useful as a high-level summary. However, the more robust and informative characterization of MLLM graphical perception is the bipartite division into a high-accuracy geometric tier

and a low-accuracy color tier.

5.3. The Distractor Hypothesis and Spatial Binding

The within-channel performance variations initially suggested that spatial distance between elements might degrade estimation accuracy. Experiment 2 appears to decouple spatial distance from continuous estimation precision. Increasing distance degraded binary identification accuracy by increasing the number of sign errors. It had no significant effect on LAE.

This might indicate that distractor elements are a primary driver of continuous degradation in accuracy in MLLMs. The model's accuracy seems to drop when visual clutter or distracting geometric shapes are introduced. This aligns with recent findings on visual attention sinks in MLLMs. Models might disproportionately allocate their attention budget to irrelevant background patches [15]. This could dilute their focus on critical data points. Distractors might fragment the model's self-attention mechanism in the context of graphical perception. Extraneous visual elements could act as attention sinks. Stripping charts of unnecessary grid lines and borders and distractors might directly improve MLLM estimation accuracy.

The distance-induced drop in binary accuracy in Experiment 2 highlights a potential vulnerability in spatial binding. The model seems to struggle to correctly bind the alphanumeric labels to their corresponding visual segments when elements move further apart. This could cause it to misidentify which bar is smaller, while paradoxically maintaining high precision in ratio estimation. Elements that need to be compared should possibly be placed as closely together as the layout permits to prevent these alphanumeric binding errors.

5.4. Prompting Effects

Our investigation into prompt sensitivity in RQ1 revealed that instructing models to generate step-by-step reasoning or CoT only resulted in a minor reduction in estimation error. Superficial framing, like role-playing or prompt duplication, appeared to have no measurable effect. CoT yielded a small improvement in absolute error. It did not alter the relative hierarchy of the visual channels.

Our multi-model evaluation could suggest that a structural hierarchy emerges even with strictly black-and-white stimuli and direct-answer prompts. The invariance of this hierarchy across prompt variations suggests that CoT has only a limited impact on the models' intrinsic visual encoding capabilities in this context. The minor performance gains from reasoning prompts might not justify the additional computational overhead for low-level perception tasks.

5.5. Capability Tiers and Architectural Confounds

The analysis of model scale and architecture in RQ3 points to the possibility of a scaling law for graphical perception. Capability might strongly predict accuracy. Frontier models appear to outperform mid-tier models. Mid-tier models, in turn, seem to outperform small models. This performance gradient is practically mirrored by the near-perfect negative correlation between API cost and estimation error. Deploying MLLMs for highly accurate automated chart reading could currently require frontier-level pricing.

MoE architectures appeared to outperform dense networks. This effect was confounded by the exceptional performance of Gemini 3.1 Pro. Gemini achieved an unprecedented mean LAE of 0.840. The architectural advantage of MoEs might vanish when Gemini is excluded from the sample. Sparse routing could theoretically allow networks to dedicate expert sub-networks to spatial reasoning. We cannot disentangle this potential from Gemini's broader proprietary training in this sample.

5.6. Limitations

Several limitations must be acknowledged. First, the evaluation is based on a sample of six models. This sample size limits the generalizability of conclusions about the capability tier and MoE architecture. Second, the stimuli utilized were strictly black-and-white and highly standardized to isolate the foundational encoding channels. Real-world visualizations contain color and legends and complex layouts that might introduce unmeasured confounding variables. Third, the study employed exclusively zero-shot prompting. Few-shot prompting or fine-tuning could potentially alter the observed hierarchies and improve absolute accuracy. Finally, the rapid evolution of MLLM capabilities means these results represent a temporal snapshot. Future generations of vision encoders might overcome the limitations observed here.

6. Conclusion

This thesis investigated whether MLLMs exhibit performance differences across visualization channels similar to those observed in human graphical perception. To answer this question, the graphical perception experiments of Heer and Bostock were replicated and evaluated with several multimodal LLMs. The original experimental setup was reproduced as closely as possible and extended with two additional tasks. A follow-up experiment further examined whether the spatial distance between the visual elements being compared affects model performance. The results show that multimodal LLMs do not perform equally across all visualization channels. Geometric channels such as position, length and angle generally produced comparable levels of accuracy across the tested models. In contrast, color-based channels, particularly luminance and saturation, resulted in substantially larger estimation errors. This finding is consistent with the established literature on human graphical perception and suggests that multimodal LLMs share at least some characteristics with human visual processing. At the same time, no clear ranking emerged among the geometric channels. While humans consistently show differences between channels, such as position and angle, the tested models achieved similar performance across most geometric encodings. This indicates that the performance profile of multimodal LLMs cannot simply be assumed to mirror the human channel hierarchy and should instead be studied directly. The additional experiment found no significant effect of distance on estimation accuracy. Increasing the separation between compared visual elements did not noticeably degrade performance. In contrast, tasks that introduced additional visual elements and competing information consistently resulted in larger errors. These findings suggest that visual distractors are more likely to be the limiting factor than spatial distance itself. From a practical perspective, the results offer several recommendations for designing systems that rely on multimodal LLMs to interpret visualizations. First, geometric encodings should be preferred whenever accurate quantitative reasoning is required. Second, important quantitative information should not rely exclusively on luminance or saturation. Third, visualizations should avoid unnecessary visual clutter and distracting elements, as these appear to have a stronger influence on model performance than the physical distance between relevant marks. Following these principles is likely to reduce interpretation errors and improve the reliability of LLM-based visualization analysis systems. Overall, the findings demonstrate that multimodal LLMs exhibit clear channel-dependent performance differences and that these differences partially align with established results from human graphical perception research. By revealing areas of convergence and divergence between human and machine graphical perception, this study not only provides a foundational step toward understanding how modern vision-language models interpret quantitative visualizations and where their current limitations lie, but also highlights the broader impact for both visualization design and AI system development. These insights

inform future efforts to optimize visualizations for mixed human-AI environments and support the advancement of more reliable, interpretable multimodal AI systems.

6.1. Outlook

Several directions for future work emerge from the findings of this thesis. A first priority is a more controlled investigation of visual distractors. While the results suggest that distractors have a stronger impact on estimation accuracy than distance, the current experiments were not specifically designed to isolate this effect. Future studies could systematically manipulate the number and type of distractors while keeping all other aspects of the stimulus constant. A design similar to the comparison between Tasks 8 and 9 could be extended to all visualization channels to determine whether distractor effects are channel-specific or represent a general limitation of multimodal LLMs. A second avenue for future research is the evaluation of a broader range of models. The landscape of vision-language models is evolving rapidly and the present study covers only a subset of currently available systems. Testing additional open-source and proprietary models would provide a more comprehensive understanding of how architectural differences influence graphical perception performance. Finally, future work should investigate the internal mechanisms that drive model behavior. In particular, studying attention patterns in vision encoders and ViTs could reveal which visual cues models rely on to extract quantitative information from charts. Human observers often use specific strategies when reading visualizations. For example, when estimating values on a bar chart, they typically focus on the bar's endpoint and the corresponding axis reference. A deeper analysis of model attention could determine whether multimodal LLMs use similar cues or whether they rely on fundamentally different visual features. Such insights would not only improve our understanding of model behavior but could also inform the design of visualizations that are easier for both humans and AI systems to interpret. Ultimately, combining behavioral experiments with analyses of internal model representations may help establish a more complete theory of graphical perception for multimodal LLMs and support the development of more reliable visualization-aware AI systems.

A. Appendix

A.1. Experimental Prompts

The following listings show the base structure of the prompt templates used in the experiments. These files are available in the project repository under `code/experiment/prompts`. As described in Section 3.4, these base templates are dynamically adapted depending on the visual encoding channel of the specific task. Specifically, generic terms like "values" or "sections" are replaced with channel-specific terminology.

- **Position & Length (Tasks 1–5):** "height" and "baseline"
- **Angle (Task 6):** "angle" and "arc"
- **Area (Tasks 7–9):** "area"
- **Color (Tasks 10–11):** "color luminance" and "color saturation"

A.1.1. No Reasoning, No Role, No Duplication

```
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {APPROACH_DETAIL}.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Is A smaller than B": true,
        "percentage": "50%"
    }
```

A.1.2. Reasoning, No Role, No Duplication

```
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Think step by step about how you will identify the marked elements,
    estimate the quantitive values of them and calculate the percentage. Write your
    reasoning process down in the specified output field.
```

```
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Reasoning": "<Write your reasoning here>",
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Reasoning": "I read the value of A. I read the value of B. I calculated the
            percentage. I gave the answer in the specified output format.",
        "Is A smaller than B": true,
        "percentage": "50%"
    }
```

A.1.3. No Reasoning, Role, No Duplication

```
Your role: You are an expert in chart interpretation, visual perception, information
    extraction and low-level chart QAs.
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {APPROACH_DETAIL}.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Is A smaller than B": true,
        "percentage": "50%"
    }
```

A.1.4. Reasoning, Role, No Duplication

```
Your role: You are an expert in chart interpretation, visual perception, information
    extraction and low-level chart QAs.
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Think step by step about how you will identify the marked elements,
    estimate the quantitive values of them and calculate the percentage. Write your
    reasoning process down in the specified output field.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Reasoning": "<Write your reasoning here>",
        "Is A smaller than B": true/false,
        "percentage": "XX%"
```

```
    }
Output Example:
    {
        "Reasoning": "I read the value of A. I read the value of B. I calculated the
            percentage. I gave the answer in the specified output format.",
        "Is A smaller than B": true,
        "percentage": "50%"
    }
```

A.1.5. No Reasoning, No Role, Duplication

```
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {APPROACH_DETAIL}.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Is A smaller than B": true,
        "percentage": "50%"
    }
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {APPROACH_DETAIL}.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Is A smaller than B": true,
        "percentage": "50%"
    }
```

A.1.6. No Reasoning, Role, Duplication

```
Your role: You are an expert in chart interpretation, visual perception, information
    extraction and low-level chart QAs.
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {APPROACH_DETAIL}.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
```

```
    {
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Is A smaller than B": true,
        "percentage": "50%"
    }
Your role: You are an expert in chart interpretation, visual perception, information
    extraction and low-level chart QAs.
Task: You are given a chart displaying quantitive attributes. Two elements in the
    chart are marked as ( A ) and ( B ). Your task is to {TASK_DESCRIPTION}.
Approach: Make a quick visual judgement to {APPROACH_DETAIL}.
Output Format: The output must be in this exact JSON format and cannot contain any
    other text:
    {
        "Is A smaller than B": true/false,
        "percentage": "XX%"
    }
Output Example:
    {
        "Is A smaller than B": true,
        "percentage": "50%"
    }
```

A.2. Stimuli Grouping Details

The following JSON document details the value pairs and ratios assigned to each of the five evaluation groups, along with the set of omitted pairs:


```
[
  {
    "group_id": 1,
    "pairs": [
      { "smaller_value": 10.000, "larger_value": 56.234, "ratio": 0.1778 },
      { "smaller_value": 10.000, "larger_value": 46.416, "ratio": 0.2154 },
      { "smaller_value": 12.115, "larger_value": 46.416, "ratio": 0.2610 },
      { "smaller_value": 17.783, "larger_value": 56.234, "ratio": 0.3162 },
      { "smaller_value": 21.544, "larger_value": 56.234, "ratio": 0.3831 },
      { "smaller_value": 17.783, "larger_value": 38.312, "ratio": 0.4642 },
      { "smaller_value": 10.000, "larger_value": 17.783, "ratio": 0.5623 },
      { "smaller_value": 26.102, "larger_value": 38.312, "ratio": 0.6813 },
      { "smaller_value": 31.623, "larger_value": 38.312, "ratio": 0.8254 }
    ]
  },
  {
    "group_id": 2,
    "pairs": [
```

```
    { "smaller_value": 10.000, "larger_value": 56.234, "ratio": 0.1778 },
    { "smaller_value": 12.115, "larger_value": 56.234, "ratio": 0.2154 },
    { "smaller_value": 14.678, "larger_value": 56.234, "ratio": 0.2610 },
    { "smaller_value": 12.115, "larger_value": 38.312, "ratio": 0.3162 },
    { "smaller_value": 12.115, "larger_value": 31.623, "ratio": 0.3831 },
    { "smaller_value": 12.115, "larger_value": 26.102, "ratio": 0.4641 },
    { "smaller_value": 21.544, "larger_value": 38.312, "ratio": 0.5623 },
    { "smaller_value": 10.000, "larger_value": 14.678, "ratio": 0.6813 },
    { "smaller_value": 21.544, "larger_value": 26.102, "ratio": 0.8254 }
  ]
},
{
  "group_id": 3,
  "pairs": [
    { "smaller_value": 10.000, "larger_value": 56.234, "ratio": 0.1778 },
    { "smaller_value": 10.000, "larger_value": 46.416, "ratio": 0.2154 },
    { "smaller_value": 14.678, "larger_value": 56.234, "ratio": 0.2610 },
    { "smaller_value": 14.678, "larger_value": 46.416, "ratio": 0.3162 },
    { "smaller_value": 10.000, "larger_value": 26.102, "ratio": 0.3831 },
    { "smaller_value": 21.544, "larger_value": 46.416, "ratio": 0.4642 },
    { "smaller_value": 17.783, "larger_value": 31.623, "ratio": 0.5623 },
    { "smaller_value": 17.783, "larger_value": 26.102, "ratio": 0.6813 },
    { "smaller_value": 12.115, "larger_value": 14.678, "ratio": 0.8254 }
  ]
},
{
  "group_id": 4,
  "pairs": [
    { "smaller_value": 10.000, "larger_value": 56.234, "ratio": 0.1778 },
    { "smaller_value": 12.115, "larger_value": 56.234, "ratio": 0.2154 },
    { "smaller_value": 10.000, "larger_value": 38.312, "ratio": 0.2610 },
    { "smaller_value": 14.678, "larger_value": 46.416, "ratio": 0.3162 },
    { "smaller_value": 14.678, "larger_value": 38.312, "ratio": 0.3831 },
    { "smaller_value": 14.678, "larger_value": 31.623, "ratio": 0.4642 },
    { "smaller_value": 12.115, "larger_value": 21.544, "ratio": 0.5623 },
    { "smaller_value": 12.115, "larger_value": 17.783, "ratio": 0.6813 },
    { "smaller_value": 46.416, "larger_value": 56.234, "ratio": 0.8254 }
  ]
},
{
  "group_id": 5,
  "pairs": [
    { "smaller_value": 10.000, "larger_value": 56.234, "ratio": 0.1778 },
    { "smaller_value": 12.115, "larger_value": 56.234, "ratio": 0.2154 },
    { "smaller_value": 14.678, "larger_value": 56.234, "ratio": 0.2610 },
    { "smaller_value": 10.000, "larger_value": 31.623, "ratio": 0.3162 },
    { "smaller_value": 17.783, "larger_value": 46.416, "ratio": 0.3831 },
    { "smaller_value": 10.000, "larger_value": 21.544, "ratio": 0.4642 },
```

```
      { "smaller_value": 26.102, "larger_value": 46.416, "ratio": 0.5623 },
      { "smaller_value": 14.678, "larger_value": 21.544, "ratio": 0.6813 },
      { "smaller_value": 38.312, "larger_value": 46.416, "ratio": 0.8254 }
    ]
  },
  {
    "group_id": "omited values",
    "pairs": [
      { "smaller_value": 26.102, "larger_value": 56.234, "ratio": 0.4642 },
      { "smaller_value": 14.678, "larger_value": 26.102, "ratio": 0.5623 },
      { "smaller_value": 31.623, "larger_value": 56.234, "ratio": 0.5623 },
      { "smaller_value": 21.544, "larger_value": 31.623, "ratio": 0.6813 },
      { "smaller_value": 31.623, "larger_value": 46.416, "ratio": 0.6813 },
      { "smaller_value": 38.312, "larger_value": 56.234, "ratio": 0.6813 },
      { "smaller_value": 10.000, "larger_value": 12.115, "ratio": 0.8254 },
      { "smaller_value": 14.678, "larger_value": 17.783, "ratio": 0.8254 },
      { "smaller_value": 17.783, "larger_value": 21.544, "ratio": 0.8254 },
      { "smaller_value": 26.102, "larger_value": 31.623, "ratio": 0.8254 }
    ]
  }
]
```

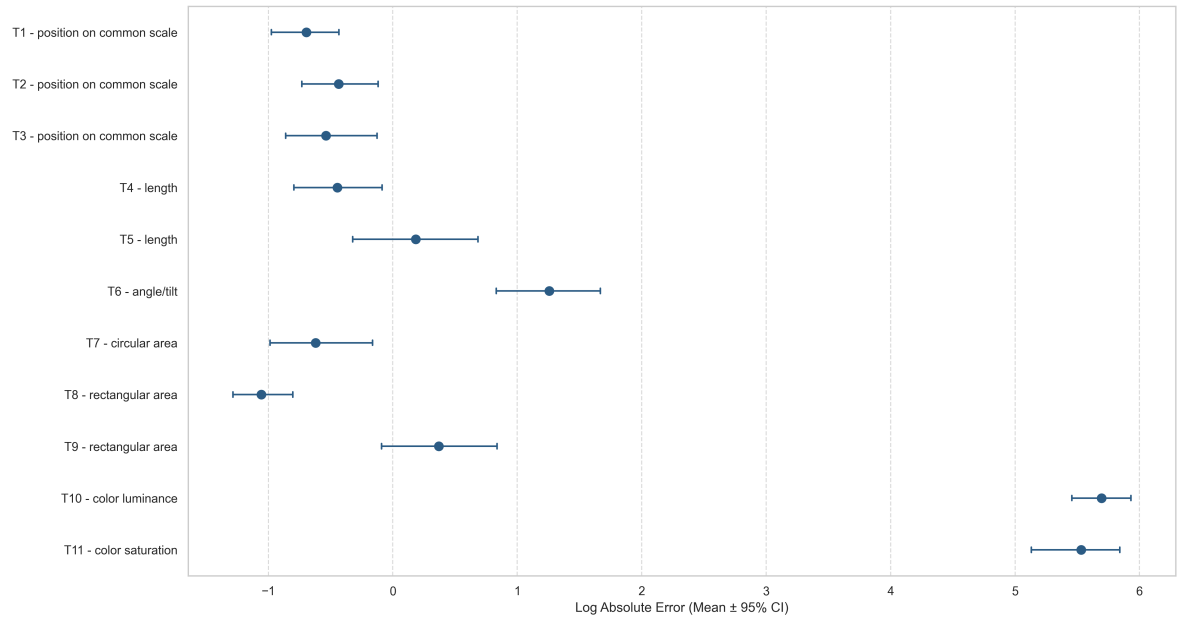

A.3. Aspect Ratio Combinations (Tasks 8 and 9)

The following JSON document details the sequence of aspect ratio pairs assigned to each rectangle in Tasks 8 and 9 across the evaluation groups:

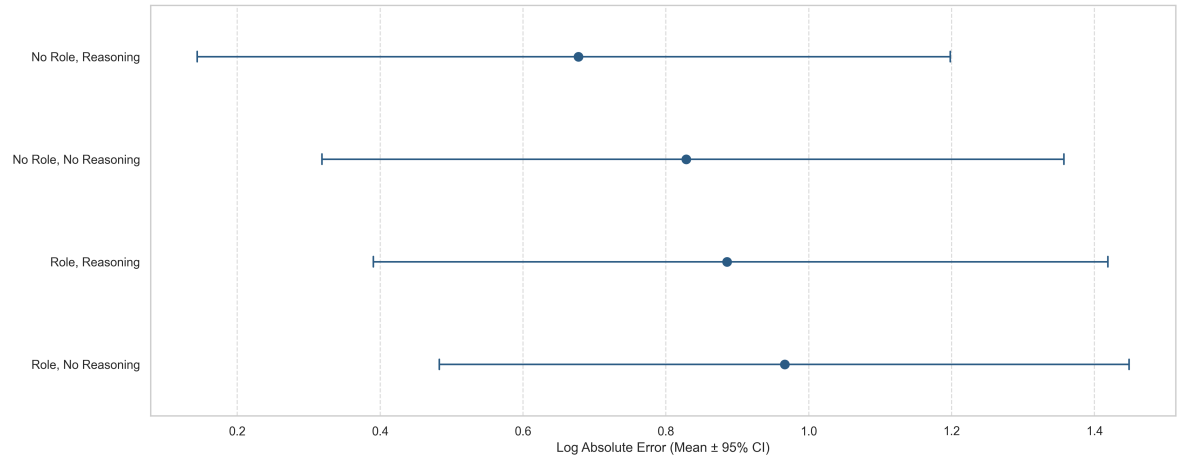

```
[
  { "index": 1, "aspect_ratio_A": "2:3", "aspect_ratio_B": "1:1" },
  { "index": 2, "aspect_ratio_A": "3:2", "aspect_ratio_B": "3:2" },
  { "index": 3, "aspect_ratio_A": "2:3", "aspect_ratio_B": "3:2" },
  { "index": 4, "aspect_ratio_A": "2:3", "aspect_ratio_B": "2:3" },
  { "index": 5, "aspect_ratio_A": "3:2", "aspect_ratio_B": "1:1" },
  { "index": 6, "aspect_ratio_A": "1:1", "aspect_ratio_B": "1:1" },
  { "index": 7, "aspect_ratio_A": "3:2", "aspect_ratio_B": "2:3" },
  { "index": 8, "aspect_ratio_A": "1:1", "aspect_ratio_B": "3:2" },
  { "index": 9, "aspect_ratio_A": "1:1", "aspect_ratio_B": "2:3" }
]
```


A.4. Extensive Per-Model Results

The following figures show the detailed per-model breakdowns of LAE distributions by task type and by prompt condition for all six evaluated models, complementing the aggregated visualizations in Chapter 4.

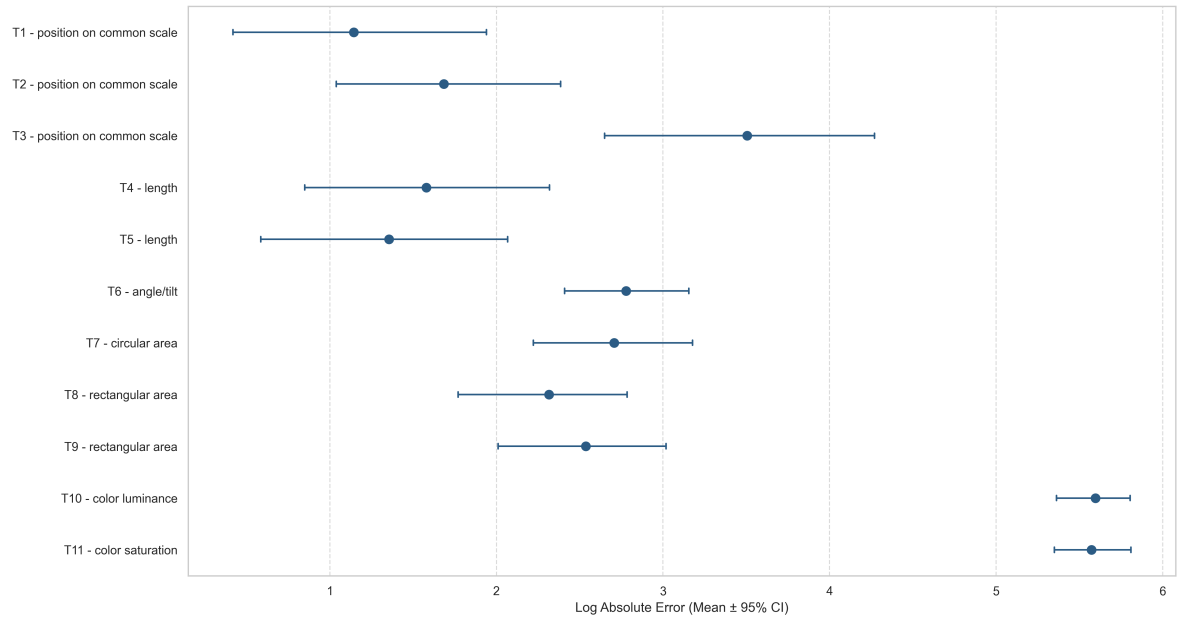


(a) Log absolute error distribution per task type

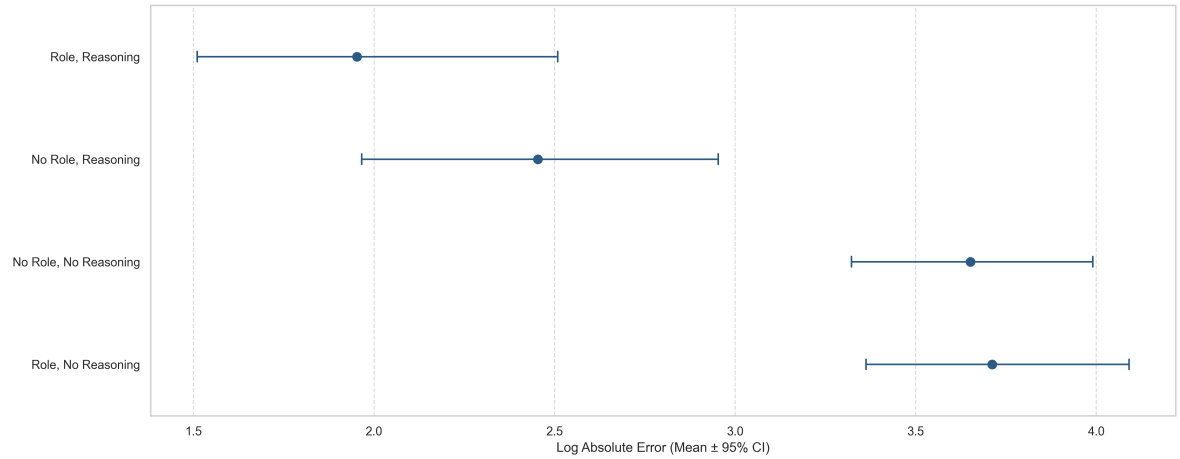


(b) Log absolute error distribution per prompt condition

Figure A.1.: Per-model LAE distributions across task types and prompt conditions for Gemini 3.1 Pro.

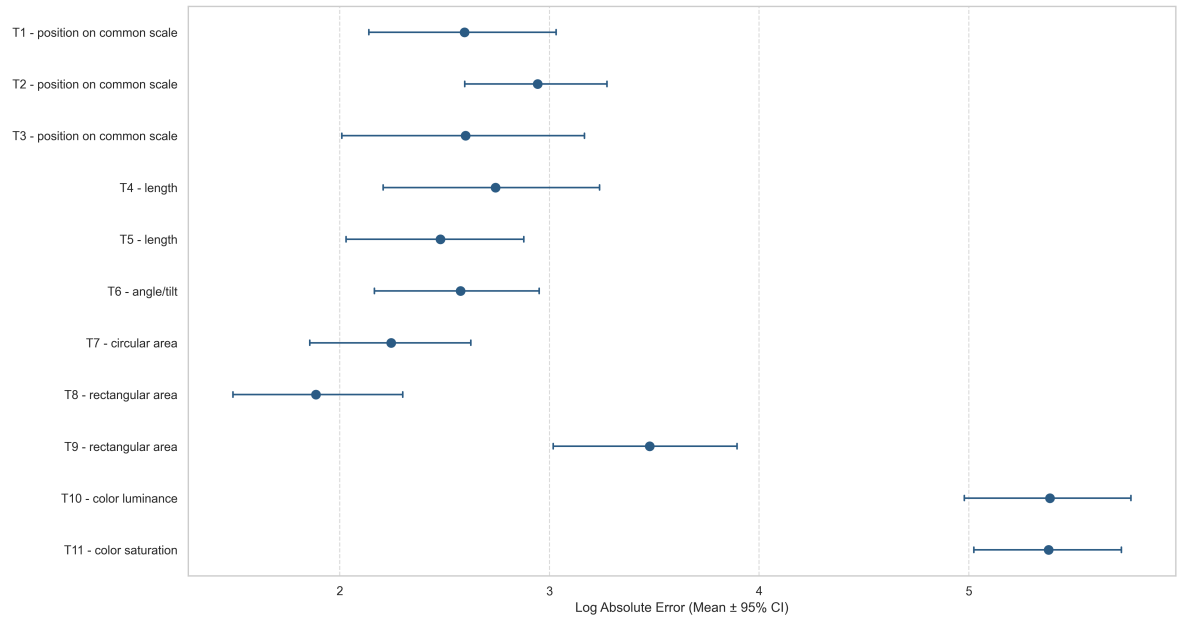


(a) Log absolute error distribution per task type

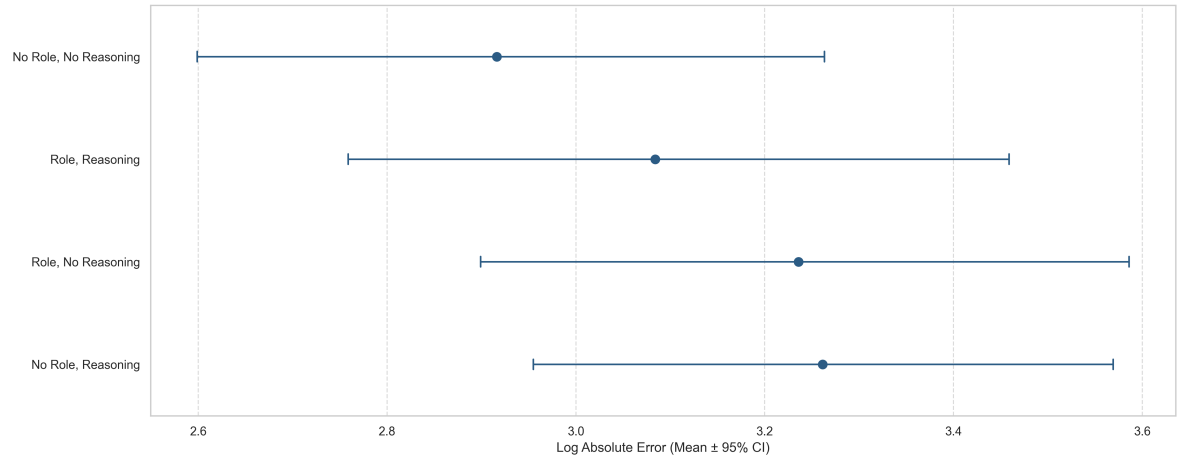


(b) Log absolute error distribution per prompt condition

Figure A.2.: Per-model LAE distributions across task types and prompt conditions for GPT-5.4.

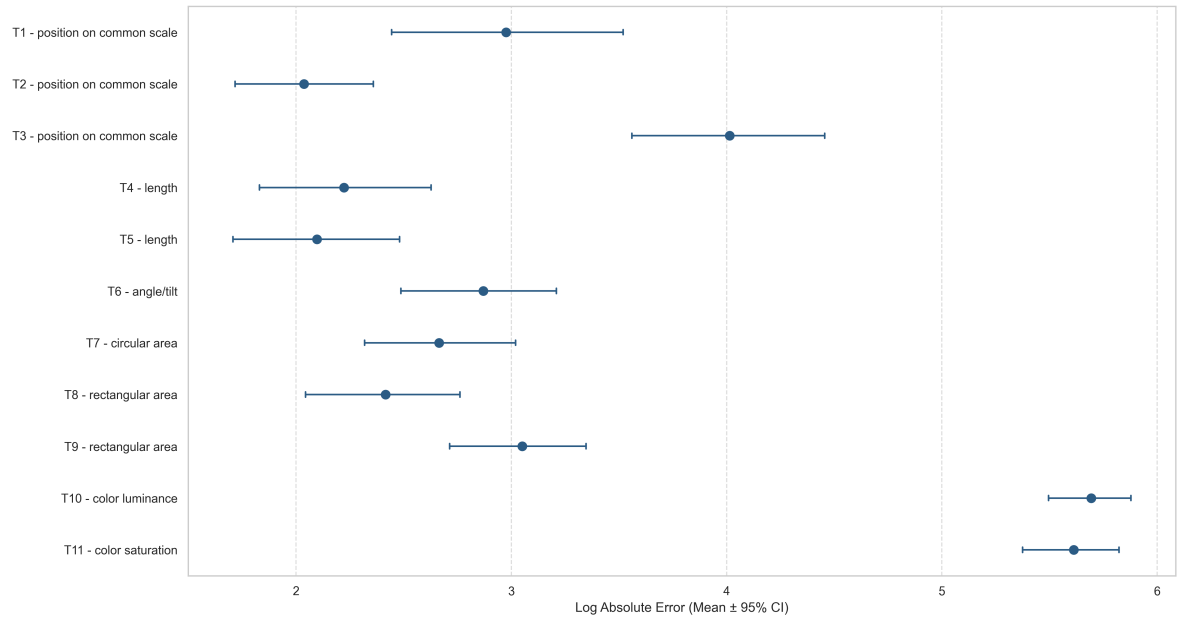


(a) Log absolute error distribution per task type

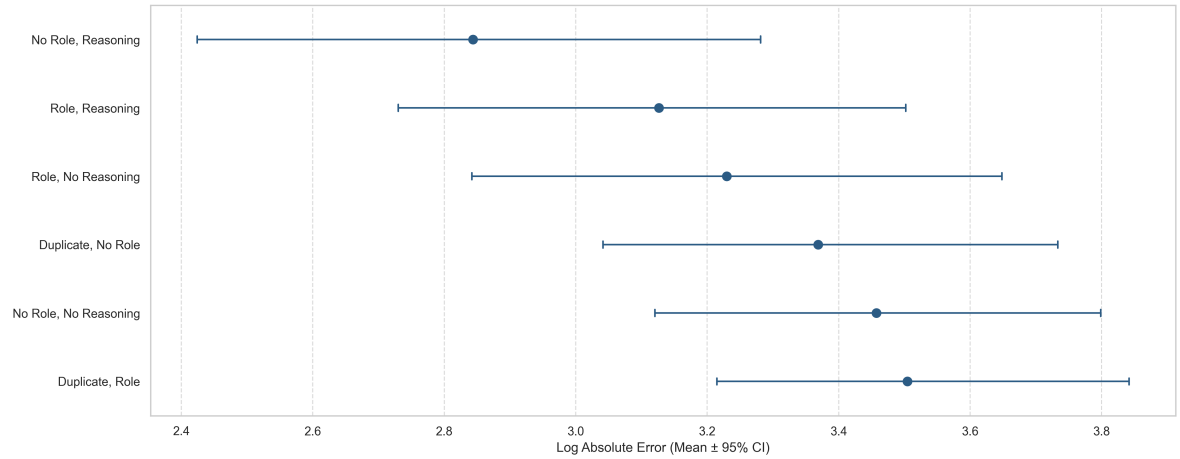


(b) Log absolute error distribution per prompt condition

Figure A.3.: Per-model LAE distributions across task types and prompt conditions for Kimi K2.6.

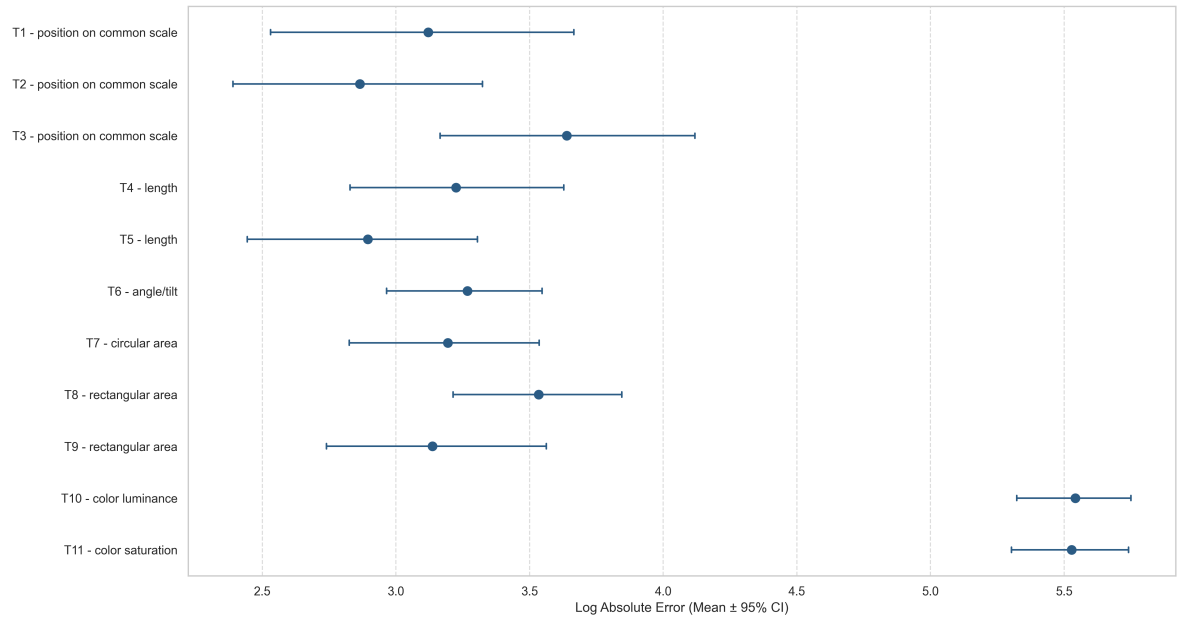


(a) Log absolute error distribution per task type

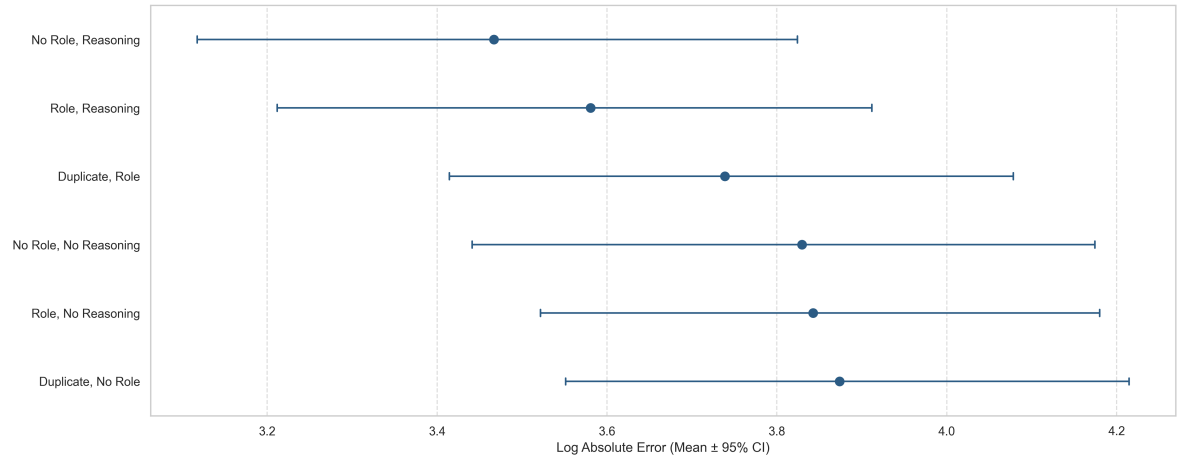


(b) Log absolute error distribution per prompt condition

Figure A.4.: Per-model LAE distributions across task types and prompt conditions for Gemma 4 31B.

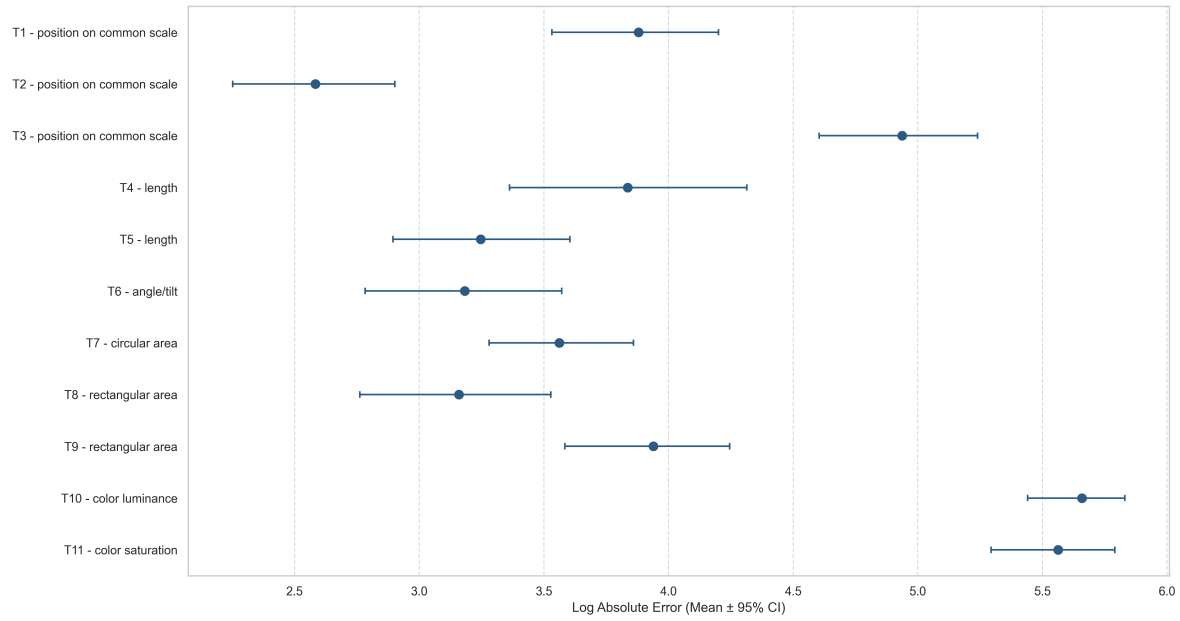


(a) Log absolute error distribution per task type

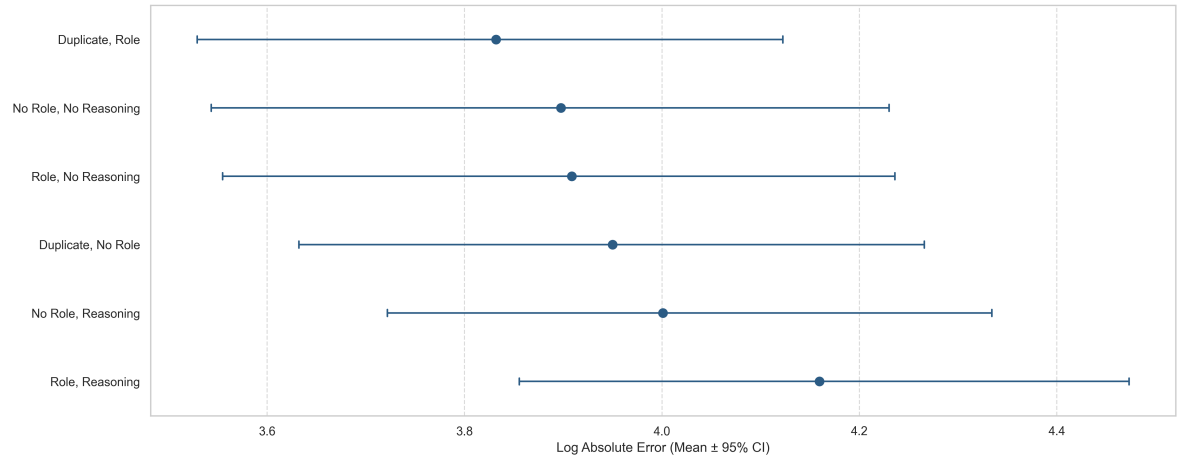


(b) Log absolute error distribution per prompt condition

Figure A.5.: Per-model LAE distributions across task types and prompt conditions for Gemma 4 26B.



(a) Log absolute error distribution per task type



(b) Log absolute error distribution per prompt condition

Figure A.6.: Per-model LAE distributions across task types and prompt conditions for Nemotron.

A.5. Supplementary Heatmaps

The following heatmaps provide additional granular data regarding the operational efficiency of the evaluated models across all task types.

A.6. Experiment 2 Binary Accuracy

A.7. Difficulty Sensitivity per Prompt Condition

Reason	Role	Model	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	Overall	Total (€)
x	x	Gemini 3.1 Pro	0.028606	0.117575	0.027061	0.037235	0.045097	0.093304	0.028958	0.032044	0.159464	0.010136	0.010824	0.053664	5.219423
		Gemma 27B	0.000033	0.000046	0.000050	0.000042	0.000045	0.000028	0.000053	0.000042	0.000048	0.000048	0.000047	0.000044	0.004328
		Gemma 31B	0.000075	0.000092	0.000080	0.000058	0.000070	0.000101	0.000083	0.000069	0.000069	0.000083	0.000074	0.000078	0.007697
		Kimi K2.6	0.016233	0.021396	0.030878	0.029847	0.017148	0.027882	0.018873	0.016119	0.057635	0.022146	0.012764	0.024629	2.379528
		Nemotron	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
		GPT-5.4	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.001651	0.163489
		Prompt Overall	***0.007913***	***0.023460***	***0.009934***	***0.011472***	***0.010669***	***0.018952***	***0.008270***	***0.008321***	***0.036478***	***0.005677***	***0.004227***	***0.013216***	***7.774464***
x	✓	Gemini 3.1 Pro	0.024306	0.040898	0.032477	0.055201	0.037143	0.081458	0.031980	0.105220	0.072632	0.003782	0.008040	0.044831	4.438246
		Gemma 27B	0.000062	0.000054	0.000047	0.000061	0.000060	0.000044	0.000039	0.000045	0.000037	0.000040	0.000047	0.000049	0.004837
		Gemma 31B	0.000134	0.000081	0.000088	0.000074	0.000074	0.000074	0.000074	0.000074	0.000074	0.000074	0.000074	0.000081	0.007905
		Kimi K2.6	0.017375	0.020551	0.039140	0.036366	0.018500	0.023210	0.013093	0.020254	0.049690	0.014337	0.006379	0.023536	2.330050
		Nemotron	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
		GPT-5.4	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.001707	0.168953
		Prompt Overall	***0.007398***	***0.010549***	***0.012243***	***0.015568***	***0.009761***	***0.017749***	***0.007816***	***0.021216***	***0.020690***	***0.003323***	***0.002708***	***0.011729***	***6.949991***
✓	x	Gemini 3.1 Pro	0.038285	0.051574	0.047950	0.052659	0.048037	0.097007	0.042680	0.034455	0.107480	0.015825	0.029358	0.051392	5.087791
		Gemma 27B	0.000133	0.000138	0.000201	0.000152	0.000173	0.000122	0.000131	0.000183	0.000225	0.000072	0.000119	0.000150	0.012993
		Gemma 31B	0.000124	0.000137	0.000152	0.000192	0.000158	0.000164	0.000162	0.000148	0.000186	0.000145	0.000136	0.000155	0.015176
		Kimi K2.6	0.018846	0.038710	0.023032	0.040577	0.028418	0.045005	0.019628	0.023446	0.102884	0.011224	0.016354	0.033466	3.289663
		Nemotron	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
		GPT-5.4	0.003080	0.003192	0.003066	0.003261	0.003250	0.003182	0.003049	0.002850	0.003317	0.002649	0.002610	0.003046	0.301551
		Prompt Overall	***0.010078***	***0.015625***	***0.012869***	***0.016758***	***0.013588***	***0.024247***	***0.012830***	***0.010118***	***0.037772***	***0.004986***	***0.008096***	***0.015179***	***8.707173***
✓	✓	Gemini 3.1 Pro	0.042014	0.052920	0.049203	0.064868	0.047971	0.100799	0.048032	0.035473	0.099162	0.016022	0.015298	0.051978	5.145871
		Gemma 27B	0.000167	0.000161	0.000328	0.000155	0.000209	0.000161	0.000149	0.000175	0.000174	0.000101	0.000125	0.000173	0.016680
		Gemma 31B	0.000145	0.000159	0.000132	0.000151	0.000188	0.000123	0.000136	0.000134	0.000146	0.000103	0.000106	0.000139	0.013718
		Kimi K2.6	0.022337	0.023399	0.032023	0.037359	0.036287	0.043607	0.020192	0.023277	0.094160	0.015151	0.015503	0.033027	3.269667
		Nemotron	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
		GPT-5.4	0.003179	0.003354	0.003218	0.003475	0.003590	0.003331	0.003290	0.003518	0.003325	0.002715	0.002686	0.003244	0.321133
		Prompt Overall	***0.011307***	***0.013332***	***0.014151***	***0.017668***	***0.014708***	***0.025136***	***0.012421***	***0.010623***	***0.032828***	***0.005682***	***0.005620***	***0.014861***	***8.767069***
Duplication	**Role**	**Model**													
✓	x	Gemma 27B	0.000081	0.000086	0.000079	0.000087	0.000124	0.000081	0.000085	0.000082	0.000077	0.000085	0.000079	0.000086	0.008510
		Gemma 31B	0.000086	0.000088	0.000128	0.000107	0.000088	0.000117	0.000108	0.000088	0.000089	0.000117	0.000089	0.000101	0.009836
		Nemotron	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
		Prompt Overall	***0.000055***	***0.000058***	***0.000069***	***0.000065***	***0.000071***	***0.000066***	***0.000065***	***0.000057***	***0.000055***	***0.000065***	***0.000056***	***0.000062***	***0.018346***
✓	✓	Gemma 27B	0.000086	0.000082	0.000080	0.000119	0.000091	0.000081	0.000084	0.000081	0.000083	0.000086	0.000084	0.000087	0.008604
		Gemma 31B	0.000090	0.000137	0.000137	0.000120	0.000125	0.000137	0.000137	0.000091	0.000092	0.000099	0.000086	0.000114	0.011273
		Nemotron	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
		Prompt Overall	***0.000059***	***0.000073***	***0.000072***	***0.000080***	***0.000072***	***0.000073***	***0.000074***	***0.000057***	***0.000059***	***0.000061***	***0.000059***	***0.000067***	***0.019877***
		Grand Total Overall	***0.007348***	***0.012606***	***0.009830***	***0.012274***	***0.009745***	***0.017188***	***0.008109***	***0.010065***	***0.025428***	***0.003961***	***0.004157***	***0.010974***	***32.236921***

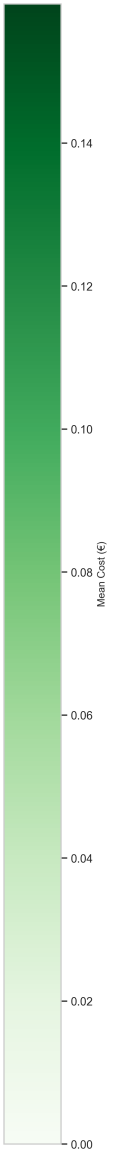


Figure A.7.: Mean API cost (in EUR) per correct binary identification across models and task types.

Reason	Role	Model	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	Overall
x	x	Gemini 3.1 Pro	2384.4	10443.2	2244.6	3166.0	3878.2	8244.8	2416.3	2695.8	14237.4	711.3	773.7	4654.2
		Gemma 27B	25.2	27.1	27.9	25.7	26.6	24.3	27.8	26.4	27.1	27.3	27.7	26.6
		Gemma 31B	31.2	31.0	31.1	31.1	30.6	30.3	30.7	30.6	31.2	31.9	32.0	31.1
		Kimi K2.6	4260.0	5664.4	8929.4	8786.9	4510.2	8172.6	4898.7	4230.4	15512.1	5868.3	3318.8	6741.1
		Nemotron	5809.9	3128.9	8125.6	1461.2	2537.8	7872.6	8627.8	952.2	1756.4	800.0	911.9	3816.7
		GPT-5.4	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0
		Prompt Overall	***2018.8***	***3219.6***	***2926.5***	***2249.0***	***1834.4***	***3901.8***	***2670.7***	***1326.4***	***5264.6***	***1243.6***	***847.8***	***2500.3***
x	✓	Gemini 3.1 Pro	1990.7	3493.6	2730.8	4789.1	3153.4	7167.4	2685.8	9319.8	6368.0	131.6	517.2	3849.8
		Gemma 27B	28.6	27.1	26.6	28.3	69.9	26.3	25.7	26.3	25.0	26.0	26.9	30.6
		Gemma 31B	86.0	31.0	31.2	31.1	31.1	31.2	30.9	31.0	31.1	32.0	32.0	36.2
		Kimi K2.6	5090.6	6310.4	12358.9	11472.1	5760.3	7266.0	3812.1	6321.0	15731.6	4429.6	1885.4	7312.5
		Nemotron	1787.1	1074.1	2340.0	1700.1	3318.6	1182.3	881.1	654.9	1188.2	584.1	706.0	1401.5
		GPT-5.4	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0
		Prompt Overall	***1527.7***	***1826.5***	***2918.4***	***3007.3***	***2035.6***	***2616.1***	***1243.1***	***2729.3***	***3894.5***	***871.0***	***531.8***	***2109.2***
✓	x	Gemini 3.1 Pro	3249.7	4453.3	4125.1	4551.7	4133.0	8568.7	3647.8	2902.8	9517.3	1215.2	2441.1	4436.9
		Gemma 27B	259.4	221.0	397.0	384.9	424.4	248.9	220.5	344.8	456.0	112.7	179.0	295.3
		Gemma 31B	160.9	181.7	208.0	219.7	204.2	169.8	163.0	202.1	210.6	101.1	109.2	175.5
		Kimi K2.6	5559.9	10907.2	6422.4	11440.7	8053.9	12193.6	5184.2	6205.5	28748.9	3418.2	5064.3	9381.7
		Nemotron	10559.9	1947.6	8760.3	2178.6	2628.4	5123.9	13162.7	713.1	5973.0	690.8	790.2	4775.3
		GPT-5.4	115.3	123.4	114.3	128.4	127.7	122.8	113.1	98.7	132.6	84.1	81.3	112.9
		Prompt Overall	***3317.5***	***2972.4***	***3451.0***	***3222.5***	***2640.4***	***4404.6***	***3518.1***	***1685.6***	***7812.9***	***937.0***	***1444.2***	***3218.7***
✓	✓	Gemini 3.1 Pro	3583.1	4571.0	4234.3	5653.2	4122.7	8907.9	4128.2	2990.7	8759.6	1228.8	1163.2	4485.7
		Gemma 27B	243.9	278.8	733.2	239.7	359.3	228.9	209.3	295.0	312.9	114.0	165.1	289.1
		Gemma 31B	180.9	200.9	182.6	230.0	208.0	157.8	194.0	177.4	219.7	93.7	110.2	177.7
		Kimi K2.6	6526.3	6187.1	8530.8	9980.8	9689.4	11678.4	5315.8	6154.0	25415.8	3945.8	4041.7	8860.5
		Nemotron	2427.4	2160.0	3880.1	2245.6	2572.7	1267.8	2373.9	811.4	3285.6	872.7	585.6	2043.9
		GPT-5.4	118.6	131.2	121.3	140.0	148.3	129.6	126.6	143.1	129.1	84.9	82.8	123.2
		Prompt Overall	***2180.0***	***2254.8***	***2947.1***	***3081.5***	***2850.1***	***3774.8***	***2129.1***	***1789.6***	***6353.8***	***1056.6***	***1024.8***	***2676.6***
Duplication	**Role**	**Model**												
✓	x	Gemma 27B	29.4	29.1	29.0	29.1	151.2	29.2	29.3	29.1	29.3	29.8	29.9	40.4
		Gemma 31B	31.1	30.8	30.9	30.8	31.0	30.6	31.0	31.0	30.8	32.0	31.8	31.1
		Nemotron	899.0	1905.1	3203.7	1311.0	4027.0	5674.6	7699.9	8044.9	2899.0	937.1	900.3	3409.2
		Prompt Overall	***319.9***	***655.0***	***1087.9***	***457.0***	***1403.1***	***1911.4***	***2586.7***	***2701.7***	***986.4***	***344.5***	***320.7***	***1161.3***
✓	✓	Gemma 27B	29.3	29.2	29.4	127.9	29.6	29.3	29.2	29.1	29.1	29.6	29.6	38.3
		Gemma 31B	31.2	31.0	31.0	31.1	31.0	31.0	31.1	31.0	30.9	32.0	32.0	31.2
		Nemotron	1585.7	2031.4	2904.1	1824.4	1259.2	2945.8	3679.4	669.4	4009.0	1418.2	870.1	2108.8
		Prompt Overall	***548.7***	***697.2***	***988.2***	***661.1***	***439.9***	***1002.0***	***1246.6***	***243.2***	***1356.3***	***493.3***	***289.0***	***724.2***
		Grand Total Overall	***1896.6***	***2189.9***	***2647.1***	***2417.9***	***2054.3***	***3223.7***	***2259.2***	***1801.6***	***4866.7***	***907.5***	***832.7***	***2281.6***

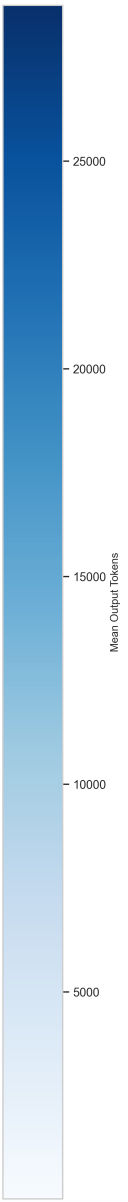


Figure A.8.: Mean generated completion tokens per response across models and task types.

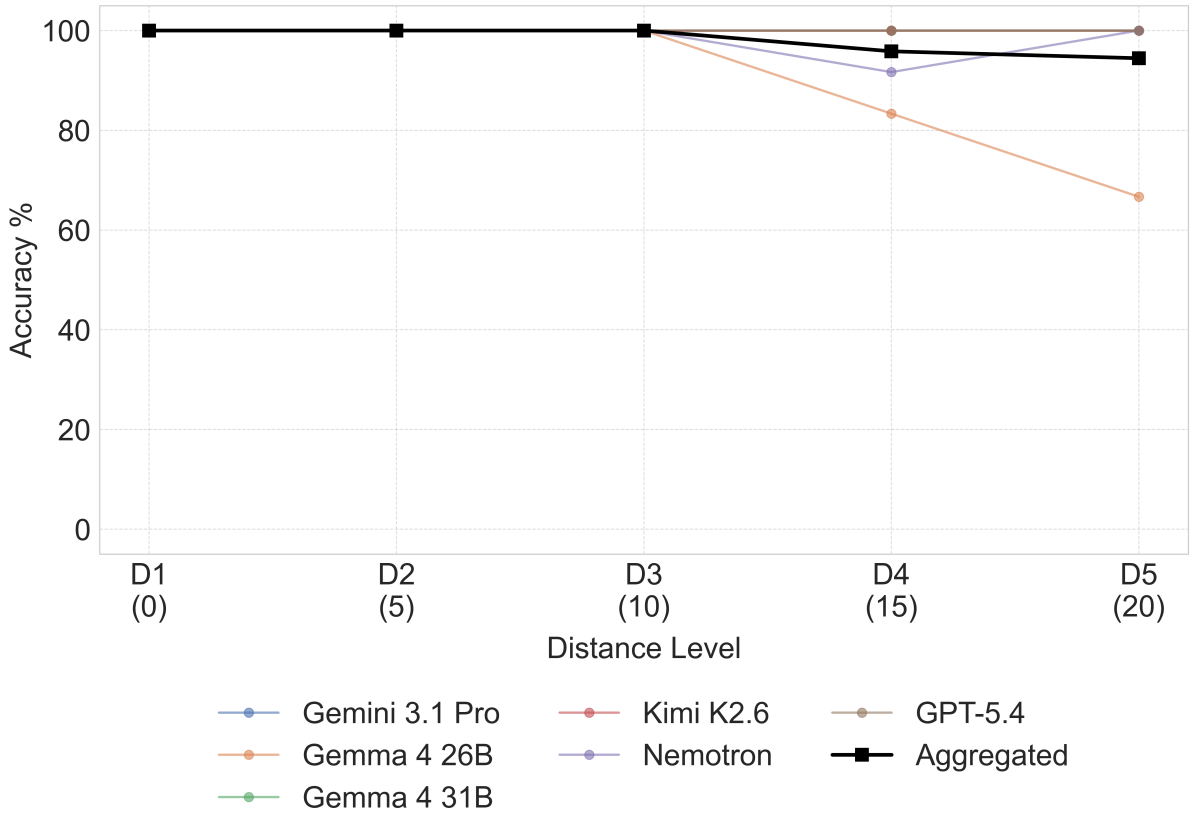


Figure A.9.: Binary identification accuracy per spatial distance level in Experiment 2.

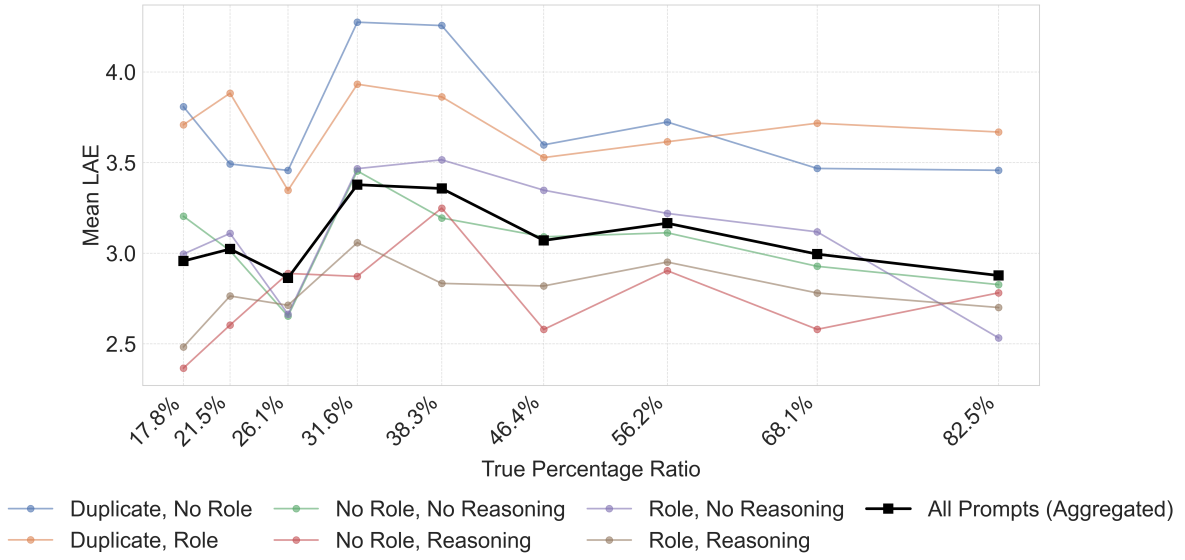


Figure A.10.: Relationship between task difficulty (true percentage ratio) and LAE per prompt condition.

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Acronyms

AI Artificial Intelligence. 1–3, 5, 24

API Application Programming Interface. 19, 27, 42, 63, 68

CLI Command Line Interface. 27

CNN Convolutional Neural Network. 2, 7

CoT Chain-of-Thought. 2, 7, 8, 22, 23, 37, 46

LAE Log Absolute Error. 28–34, 36, 37, 39, 41–47, 55–61, 65–69

LLM Large Language Model. 2, 3, 6, 21, 27

MLLM Multimodal Large Language Model. iv–vi, 1–13, 19, 22, 23, 25, 26, 28, 32, 34, 35, 37, 42, 44–48, 66, 67, 69

MoE Mixture of Experts. 3, 6, 20, 21, 40, 47, 69

ViT Vision Transformer. 2, 6, 7, 11, 26, 49

Bibliography

[1] W. S. Cleveland and R. McGill. "Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods". In: *Journal of the American Statistical Association* 79.387 (1984), pp. 531–554. DOI: 10.1080/01621459.1984.10478080.

[2] J. Heer and M. Bostock. "Crowdsourcing Graphical Perception Using Mechanical Turk to Assess Visualization Design". In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2010, pp. 203–212. DOI: 10.1145/1753326.1753357.

[3] C. Ware. *Information Visualization: Perception for Design*. 2nd ed. San Francisco: Morgan Kaufmann, 2004.

[4] J. Bertin. *Semiology of Graphics: Diagrams, Networks, Maps*. University of Wisconsin Press, 1983.

[5] J. D. Mackinlay. "Automating the Design of Graphical Presentations of Relational Information". In: *ACM Transactions on Graphics* 5.2 (1986), pp. 110–141. DOI: 10.1145/22949.22950.

[6] P. Poonam, P.-P. Vázquez, and T. Ropinski. "Evaluating Graphical Perception Capabilities of Vision Transformers". In: *Computers & Graphics* (2026). arXiv:2602.18178.

[7] M. Sinnona, V. Bonas, V. Siless, and E. Iarussi. "Do Large Language Models Understand Data Visualization Principles?" In: *arXiv preprint* (2026). arXiv:2602.20084.

[8] Y. Han, C. Zhang, X. Chen, X. Yang, Z. Wang, G. Yu, B. Fu, and H. Zhang. "ChartInsights: Evaluating Multimodal Large Language Models for Low-Level Chart Question Answering". In: *Proceedings of Empirical Methods in Natural Language Processing (EMNLP)*. 2024.

[9] T. Munzner. *Visualization Analysis and Design*. A K Peters Visualization Series. CRC Press, 2014. ISBN: 9781466508910.

[10] B. B. Bederson, B. Shneiderman, and M. Wattenberg. "Ordered and Quantum Treemaps: Making Effective Use of 2D Space to Display Hierarchies". In: *ACM Transactions on Graphics (TOG)* 21.4 (2002), pp. 833–854. DOI: 10.1145/571647.571649.

[11] G. Guo, J. J. Kang, R. S. Shah, H. Pfister, and S. Varma. "Understanding Graphical Perception in Data Visualization through Zero-shot Prompting of Vision-Language Models". In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2411.00257. 2024.

[12] H. Liu, C. Li, Q. Wu, and Y. J. Lee. "Visual Instruction Tuning". In: *Advances in Neural Information Processing Systems*. 2023.

[13] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale". In: *International Conference on Learning Representations (ICLR)* (2021). arXiv:2010.11929.

[14] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei. "Scaling Laws for Neural Language Models". In: *arXiv preprint arXiv:2001.08361* (2020).

[15] L. Dong and A. Crisan. "Evaluating LLMs for Visualization Tasks". In: *arXiv preprint* (2024). Evaluates GPT-4o, GPT-3.5, Gemini, and GPT-5.4 on chart generation and understanding tasks.

[16] L. Dong and A. Crisan. "Do MLLMs See What We See? Analyzing Visualization Literacy Barriers". In: *arXiv preprint* (2025). arXiv:2504.05445.

[17] Anonymous. "Understanding Bias in Large-Scale Visual Datasets". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Framework for dissecting dataset biases via image transformations and language models. 2024.

[18] S. Schulhoff, M. Ilie, N. Balepur, K. Kahadze, A. Liu, C. Si, Y. Li, A. Gupta, H. Han, S. Schulhoff, P. S. Dulepet, S. Vidyadhara, D. Ki, S. Agrawal, C. Pham, G. Kroiz, F. Li, H. Tao, A. Srivastava, H. D. Costa, S. Gupta, M. L. Rogers, I. Goncearenco, G. Sarli, I. Galynker, D. Peskoff, M. Carpuat, J. White, S. Anadkat, A. Hoyle, and P. Resnik. "The Prompt Report: A Systematic Survey of Prompt Engineering Techniques". In: *arXiv preprint arXiv:2406.06608* (2024).

[19] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models". In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2201.11903. 2022.

[20] Anonymous. "Prompt Repetition Improves Non-Reasoning LLMs". In: *arXiv preprint arXiv:2512.14982* (2025).

[21] E. Erdil. *How do mixture-of-experts models compare to dense models in inference?* Accessed: 2026-05-19. 2024. URL: `https://epoch.ai/gradient-updates/moe-vs-dense-models-inference`.

[22] B. Xu, A. Yang, J. Lin, Q. Wang, C. Zhou, Y. Zhang, and Z. Mao. "ExpertPrompting: Instructing Large Language Models to be Distinguished Experts". In: *arXiv preprint arXiv:2305.14688* (2023).